



LoD

The logic of Descriptions

(HP2T)

LoD – The Logic of descriptions

- **Intuition**
- Definition
- Domain
- Language
 - Etype percepts
 - Composite etype percepts
 - Descriptions / definitions
- Interpretation function
 - Etype percepts
 - Composite etype percepts
 - Descriptions / definitions
- Theories
 - Teleontologies
 - Teleologies
- Entailment
- Reasoning problems
 - Satisfiability
 - Subsumption
 - Equivalence
 - Disjointness
 - Reduction to Satisfiability
- Key notions

LoD – Why a logic of Descriptions?

- The **Logic of Descriptions** (LoD) allows us to represent etypes and dtypes and the properties which correlate them. It allows to reason about the etypes and dtypes which are populated in a LoE EG.
- It allows to construct a new **complex etype** based on the **properties** of the elements of an existing etype. For instance a foreigner may be described as a person who does not speak the local language, or a pet as an animal which lives in the house
- It allows to construct a new **complex composite etype** from simpler etypes, starting from the basic ones, as defined in LoE. For instance, it allows to define a parent as the union of father and mother.
- It allows to **constrain** the extension of etypes (via a **description**). For instance, it allows to say that a woman and a man are disjoint etypes, and that they are more specific than the etype person.
- It allows to **define** a new type by giving a name (via a **definition**) to a previously constructed complex etype
- It is the logic for reasoning about **ETGs**.

LoD – Highlights

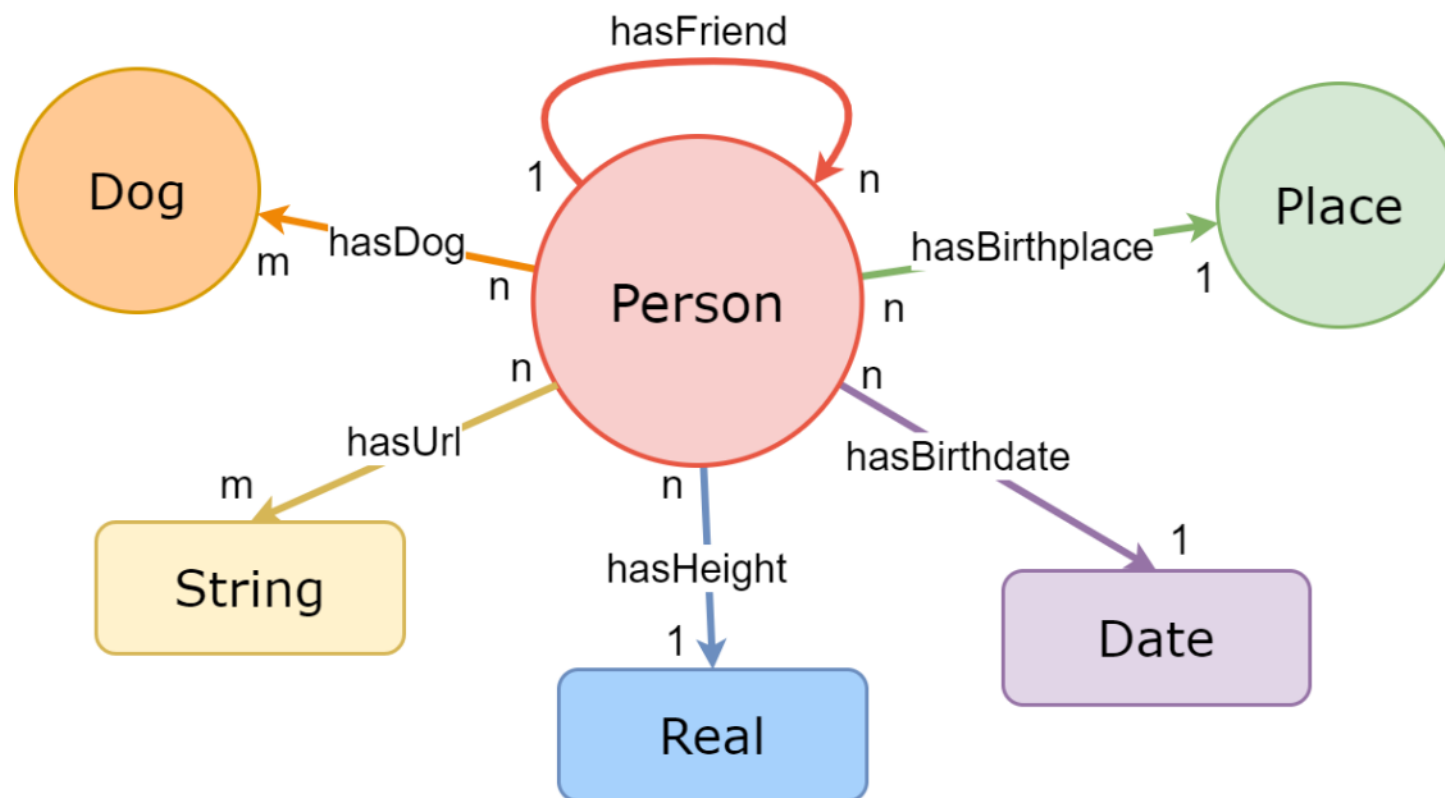
- LoD is the logic encoding the **general knowledge** about entities (how they are defined and how they can be described).
- LoD represents etypes and the properties and relations of etypes. It does not represent specific entities. Knowledge is at the type level, not at the entity level.
- LoD is a world logic with a graph linguistic/ analogic representation.
- Any LoE EG is built with reference to a **reference schema** extracted from a **LoD TLO**. A TLO can be thought of as specifying the schema of EGs.
- The LoD language **substantially extends** the language used to write ETGs allowing for complex etypes, negation and disjunction.
- LoD is conceptually similar to the Terminology Box (TBox) of Description Logics (DL). The move is from DBs to KGs.

LoD – which (names of) percepts

In LoD we have the following ETG elements:

- An **entity type** (etype) is a class of entities (corresponding to the etype to which an entity belongs in a LoE EG).
- A **datatype** (dtype) is a class of (data) values (corresponding to the dtype to which a value belongs in a LoE EG).
- An **Object Property** describes a relation between two etypes (not between two entities, as in LoE).
- A **Data Property**, also called **Attribute**, describes a characteristic of an etype (not of an entity as in LoE).

An example of LoD ETG



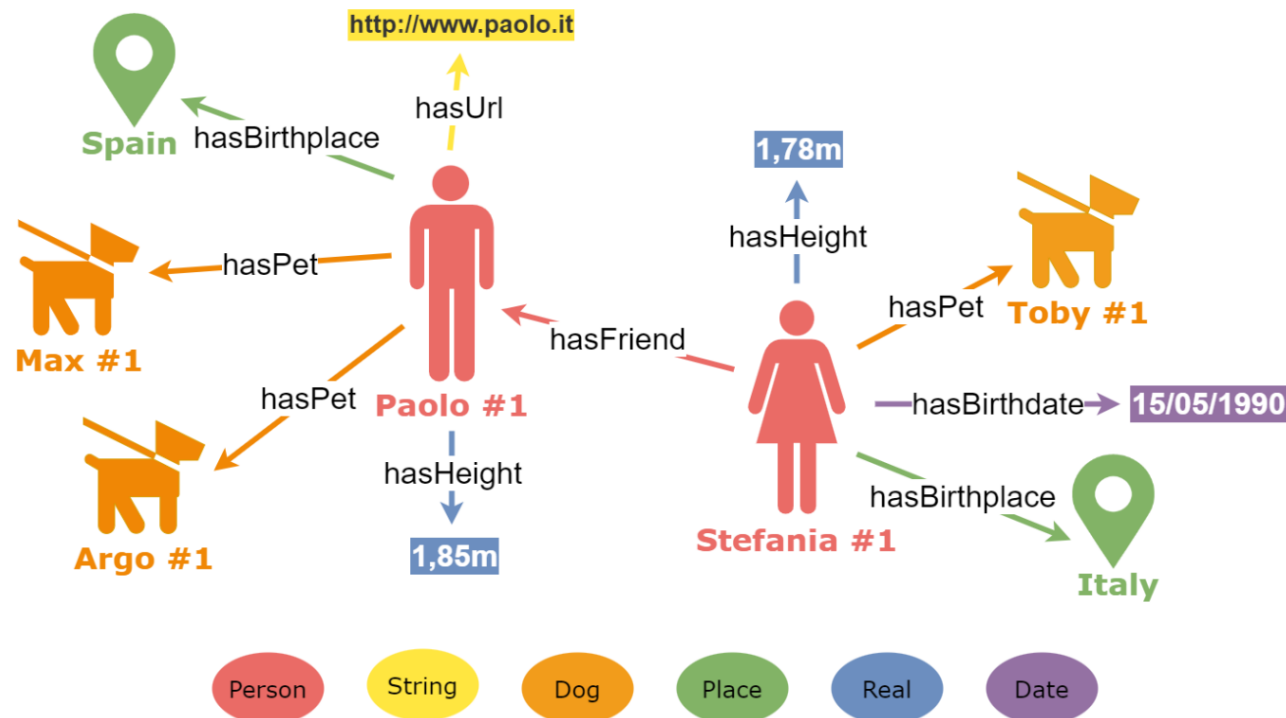
Which percepts?
Which facts?

Well-formedness conditions

An ETG, to be well-formed must satisfy the following conditions:

- Each node is associated one and ONLY one etype/ dtype.
- Each link is associated with one and only one data or object property.
- Data and object properties must have the correct etypes or datatypes (strong typing).
- No links are allowed starting from dtype nodes.

An example of EG for the previous ETG



Observation (ETG, EG). An ETG defines all the etypes, dtypes, object properties and attributes used in an EG. An EG is an expansion (notion formally defined later on) where each node and link is expanded into all its elements. Compare with the previous ETG.

LoD – which alphabet elements?

The **same** alphabet elements as percepts, that is:

- **Entity, etype, value, dtype, Attribute, Object Property** (as from ETG above).

plus language elements needed to build knowledge statements

- Etype specializations, using object and data properties
- Etype constructors, that is: intersection, union, complement.
- Defined etypes
- Equivalence/ subsumption /disjointness assertions of etypes (Semantically: set equality, subset relation, disjointness)

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LoD – The Logic of Descriptions - definition

We formally define LoD as follows

$$\mathbf{LoD} = \langle \mathbf{ETG}, \models_{\mathbf{LoD}} \rangle$$

with

$$\mathbf{ETG} = \langle \mathbf{L}_{\mathbf{LoD}}, \mathbf{D}, \mathbf{I}_{\mathbf{LoD}} \rangle$$

When no confusion arises, we drop the subscripts.

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Domain

Definition (Domain)

$$D = \langle \{C\}, \{R\} \rangle$$

where:

- $\{u\}$ is a set of **units** $u_1, \dots, u_n$, where $u_1 \in U$, with $U = \{u\}$ is the **universe** of D
- $\{C\}$ is a set of **classes** $C_1, \dots, C_m$ of units, for some m , with $C_i \subseteq U$
- $\{R\}$ is a set of **binary relations** $R_1, \dots, R_p$ between units, for some p , with $R_i \subseteq U \times U$

Observation (Domain of interpretation). The units in U are not part of the domain of interpretation of LoD. The alphabet does not mention them.

Classes

Definition (Class)

$$\{C\} = E_T \cup D_T$$

where:

- $U = \{u\}$ is the **universe of interpretation**;
- $E = \{e\} \subseteq \{u\} = U$ is the **entity universe**;
- $V = \{v\} \subseteq \{u\} = U$ is the **value universe**;
- $\{e\}$ and $\{v\}$ are disjoint.
- $E_T = \{E_i\}$ is a **set of etypes** E_i , with $E_i = \{e\} \subseteq E$
- $D_T = \{D_i\}$ is a **set of dtypes** D_i , with $D_i = \{v\} \subseteq V$

Observation (LoD classes). The same classes as LoE, but with no reference to the entities e and values v in U .

Binary relations

Definition (Binary relation)

$$\{R\} = O_R \cup A_R$$

where:

- $O_R = \{O_i\}$ is a **set of object properties** O_i , with $O_i \subseteq E_k \times E_j$
- $A_R = \{A_i\}$ is a **set of attributes** A_i , with $A_i \subseteq E_k \times D_j$

Observation (Object and data relations). The same relations as LoE.

An example of domain of ETG (continued)

$$E_T = \{\text{Entity, P, D, L, ...}\}$$

$$D_T = \{\text{Data, Real, String, ...}\}$$

$$\{R\} = \{hF, hD, hH, hB, hL, hU, ... \}$$

from which we construct the following facts in the domain:

$$D = \{P \subseteq \text{entity, Real} \subseteq \text{data, } hF(P, P), hD(P, D), hH(P, \text{Real}), ... \}$$

with, e.g., $hF(P, P)$ standing for $hF \subseteq P \times P$

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Language

Intuition (Language). The LoD language is progressively and compositionally extended in three steps:

- **Step 1 – The language of etypes (etype percepts).** It is an etype construction language. Composes etypes and properties to generate new etypes (e.g., a “mother” as a “parent” who is a “female”);
- **Step 2 – The language of composite etypes (composite etype percepts).** It is a composite etype construction language. Composes pairs of etypes to generate complex new etypes (e.g., the union of “mother” and “father”, i.e. the set denoted by “mother or father”);
- **Step 3 – The language of Assertions (LoD language).** It extends the alphabet by introducing names for defined etypes, as constructed in steps 1 and 2 (e.g., the new etype “parent” defined as “a parent is a mother or father”). Circular definitions are allowed (e.g., “parent”).

Observation (Language). The LoD language is constructed in step 3 by exploiting the language extensions of the first two steps.

Language - Observations

Observation (Percepts and facts). Step 1 and step 2 only allow for the extension of the types of etypes (semantically, percepts), which can be talked about. Step 3 allows for assertions about what is the case in the domain.

Observation (LoD assertions). The LoD language (step 3) assertions do NOT necessarily describe single facts but complex compositions of assertions (e.g., the fact that cars are vehicles and that they have four vehicles).

Observation (LoD assertions). The LoD language (step 3) allows to **define** or **describe new etypes**. Definitions introduce new terms which extend the language and, therefore, generate in the domain of interpretation new simple percepts, i.e., etypes. This is a basic mechanism by which natural language and knowledge works. Thus, for instance:

- I can **define** a car as a street vehicle which is NOT a moto-bike, or a bus, or a truck (as in the WordNet Genus-Differentia definitions).
- I can **describe** a sport car as a car which have a certain shape and goes fast.

Language – Observations

Observation (Facts, percepts). Facts (i.e., perceptions, compositions of percepts) define what is the case in a model, as they occur in the reference domain (see lectures before). These percepts and facts are not decomposable in simpler percepts and facts, respectively. We call them **atomic percepts** and **atomic facts**.

Observation (Complex percepts and facts). We distinguish the percepts and facts denoted by the LoD language between **(atomic) percepts** and **facts** and **complex percepts** and **facts**, i.e., combinations of them.

Observation (Etype and composite complex percepts). We distinguish **complex percepts** into (complex) **etypic percepts**, obtained by defining a new etype based on its properties, and (complex) **composite etypic percepts**, obtained by composing etypes.

Observation (Language and domains). The LoD language allows for the construction of percepts (step 1, 2) and facts (step 3). It creates a linguistic mental representation reality which does not map into the “reality” of what is being perceived, i.e. the analogic mental representation. But it does map to the domain (as defined in set-theory)

Observation (Language and domains). The possibility of giving a name to the newly defined etypes allows for the generation of an unbound number of etypes and facts.

Observation (Language heterogeneity). LoD enables an expressiveness which approximates the complexity of human natural language.

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Step 1 – The language of etype percepts

Definition (Language of etype percepts, L_T)

$$L_T = \langle A_T, FR_T \rangle = \{p_T\}$$

where:

- L_T is a **language** naming atomic type percepts
- A_T is an **alphabet** for etype percept generation
- FR_T is a **set of formation rules**
- $\{p_T\}$ is the set of type percepts p_T (dtypes plus atomic and complex etypes) obtained by the exhaustive application of FR_T to A_T (the transitive closure $FR_T^+(A_T)$ of FR_T applied to A_T).

World models as etype graphs (Recall)

Etype graphs are linguistic representations of analogical representations which **describe** etypes. They are **graphs** where

- Nodes are etypes (as in «married person»)
- There is any number of link denoting **object** and **data properties**, (as in «friend of», «has friend», «has height», the last cannot be reversed)

where: etypes and properties denote subsets of the percepts and pairs or percepts, respectively, as denoted by a **reference concept** which defines their universe of interpretation

NOTE: etype graphs may or may not have «**more/less general (MG/LG) etypes**» or **etype disjointness**» relations denoting, respectively, subset and empty intersection.

Etype graphs (Recall)

Etype graphs take the form of labeled directed graphs where, semantically

- The root node (etype) is an etype which has the the same extension of a reference concept which defines the Universe of interpretation for the type model (as in «person»)
- An etype which is not the root etype can be optionally described by a **description** which associates a set of properties to a **MG etype**, thus making it a **LG etype** (as in «a bachelor is a married person»)
- An etype which is not the root etype can be described as having a LG/MG relation with a **MG/LG etype** parent node, thus making it a **LG/MG etype** (as in «a bachelor is a person»»)

Alphabet

Definition (Alphabet A)*

$$A_T = \langle \{T\}, \{P\} \rangle$$

where:

- $\{T\} = \{E_i\} \cup \{D_i\}$ is a set of unary predicates standing for **etypes** and **dtypes**;
- $\{P\} = \{O_i\} \cup \{A_i\}$ is a set of binary **properties**, where O_i is an **object property**, also called a **role**, and A_i is an **attribute**.

Observation (Alphabet of percepts). Similarly to LoE, A_T is an alphabet which denotes percepts in the domain (but denoting a different set of percepts).

*The elements of the alphabet are written in *italic* to distinguish them from percepts

Formation rules – BNF

$$\langle p_T \rangle ::= \langle \text{etype} \rangle \mid \langle \text{dtype} \rangle \mid \top \mid \perp$$
$$\begin{aligned} \langle \text{etype} \rangle &::= \exists \langle \text{objProp} \rangle . \langle \text{etype} \rangle \mid \\ &\quad \exists \langle \text{dataProp} \rangle . \langle \text{dtype} \rangle \mid \\ &\quad \forall \langle \text{objProp} \rangle . \langle \text{etype} \rangle \mid \\ &\quad \forall \langle \text{dataProp} \rangle . \langle \text{dtype} \rangle \end{aligned}$$
$$\langle \text{etype} \rangle ::= E_1 \mid \dots \mid E_n$$
$$\langle \text{dtype} \rangle ::= D_1 \mid \dots \mid D_n$$
$$\langle \text{objProp} \rangle ::= O_1 \mid \dots \mid O_n$$
$$\langle \text{dataProp} \rangle ::= A_1 \mid \dots \mid A_n$$

where: $E_1, \dots, E_n, \dots, A_1, \dots, A_n$ are etypes which have the same extension as reference concepts.

Formation rules – BNF

Observation (BNF). This BNF does allow the iterative application of the formation rules on etypes (dtypes cannot be changed). It allows for the generation of etype percepts of any depth.

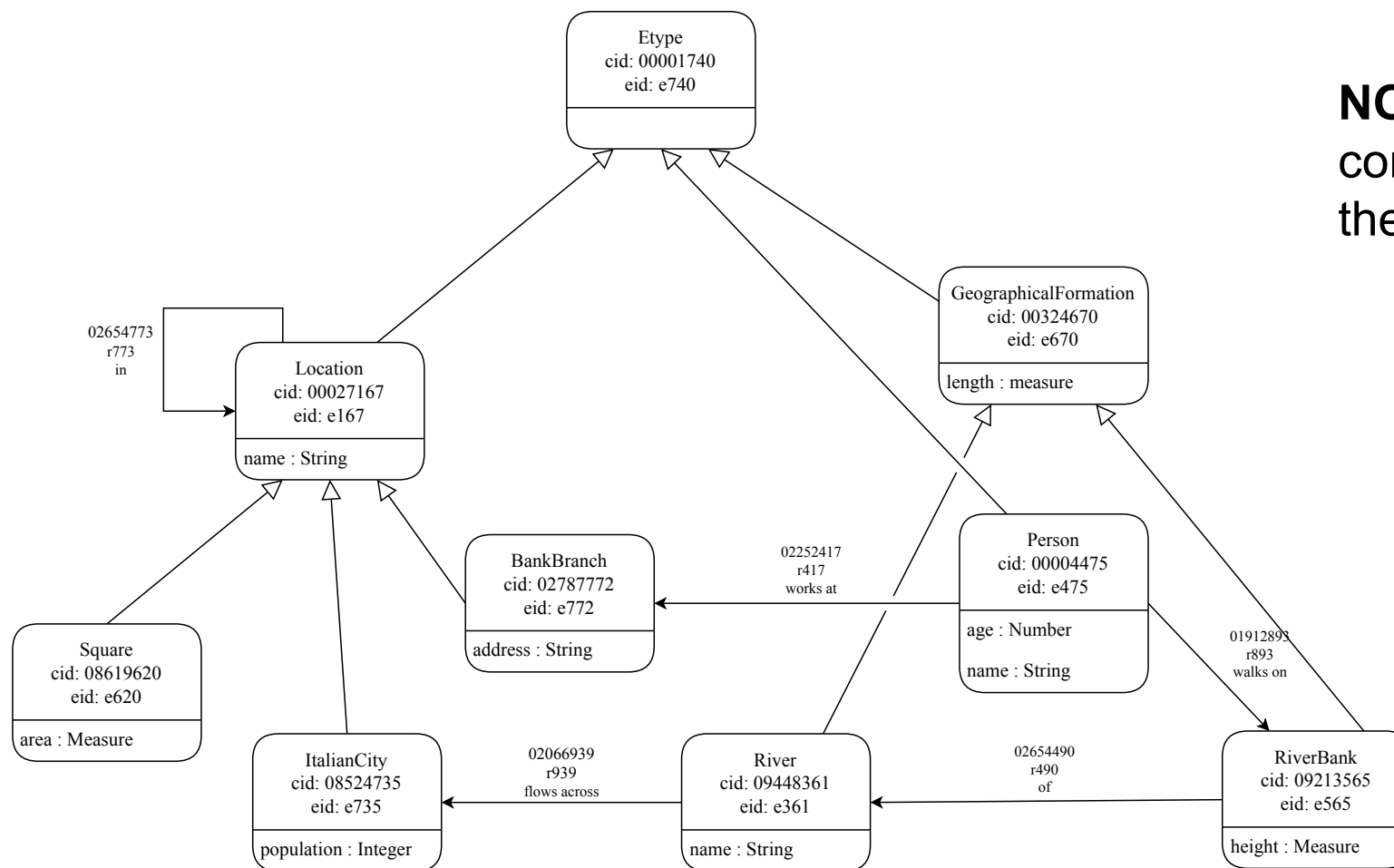
Observation (BNF). Entities are not mentioned (not part of the language). They are referred implicitly via the existential quantifier and also, somehow via the universal quantifier.

Observation (Etypes). Etype percepts which are etypes and dtypes are defined either (1) to have the same extension as their reference concepts or (2) reference concepts associated with one or more properties

Observation (Object and data properties). Etype percepts which are object and data properties are defined to have the same extension as their reference concepts

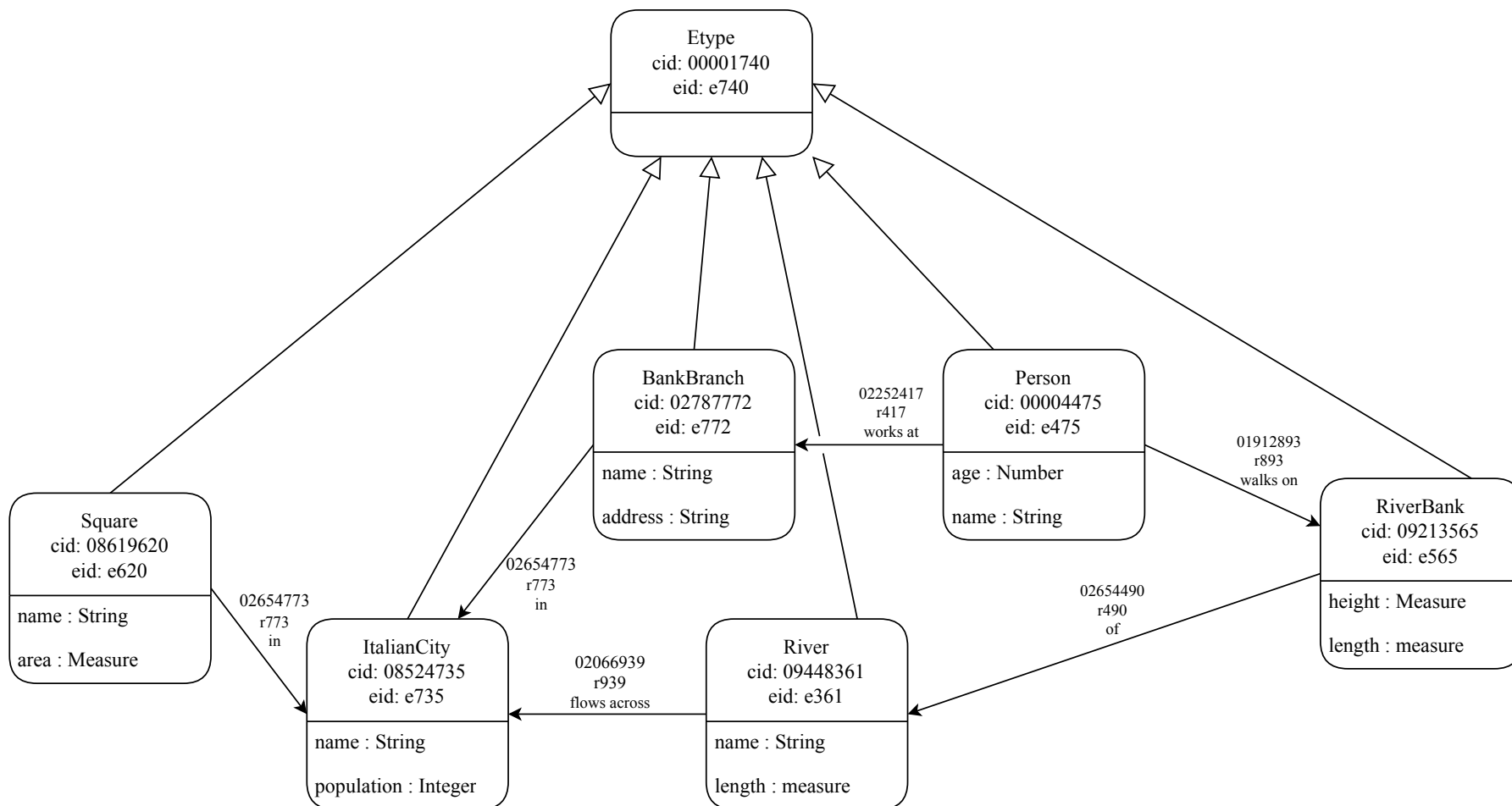
Observation (Etype percepts). Etype percepts formalize ETG graphs providing high expressibility in the definition of etypes (usually never used in mainstream ETG graphs)

Knowledge teleontology (KTLO) – example (Recall)



NOTE: Cid's of types connect to the Cid of the reference concept

Teleology (TLO) – example (Recall)



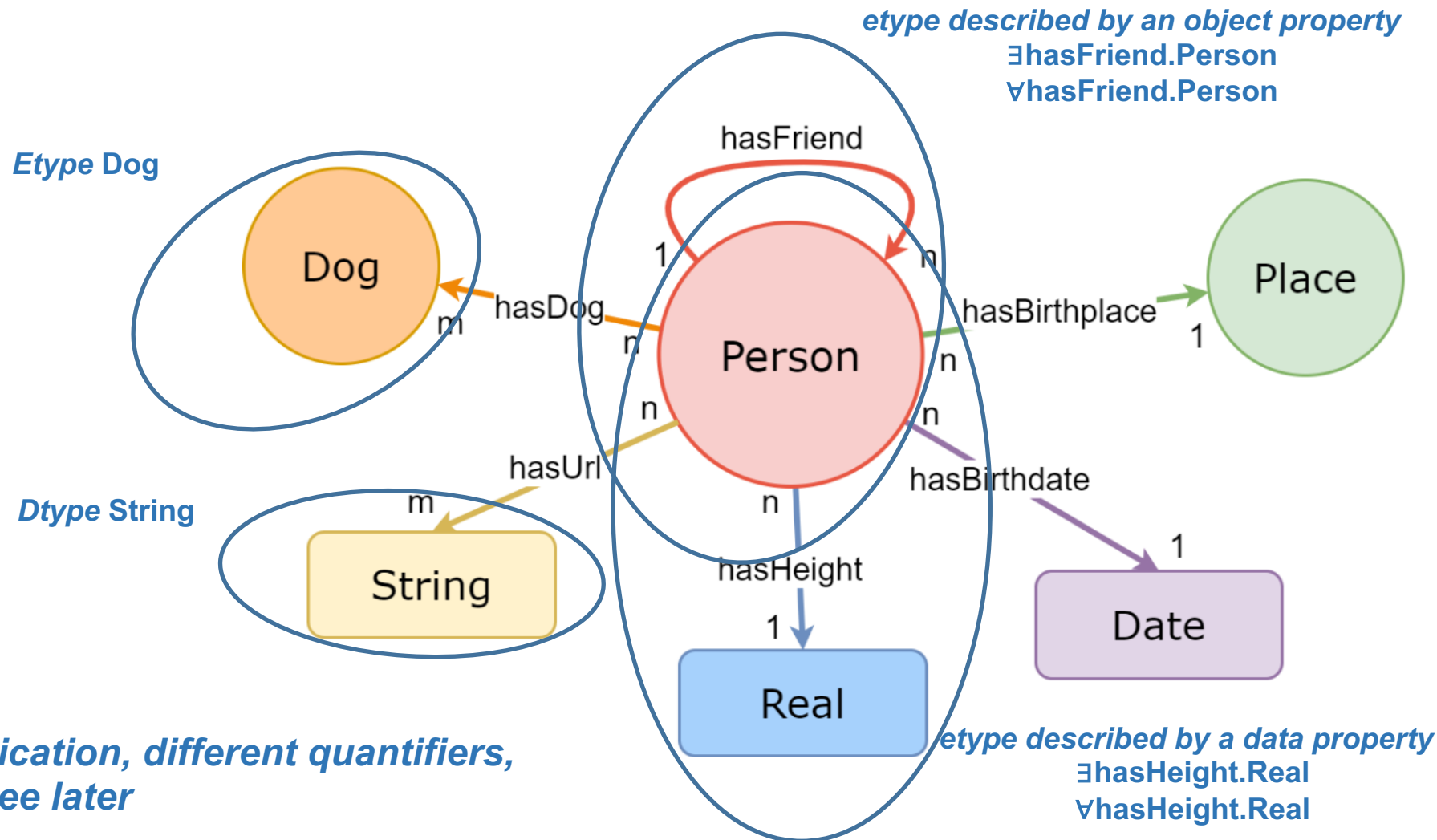
Etype percepts – Example

- $\top$ - to be read **Top** (in Concept Graphs, it is usually called **Entity**)
 - (*Intuition: the set of all entities*)
- $\perp$ - to be read **Bottom**
 - (*Intuition: the empty set of entities*)
- Person
 - (*Intuition: the set of entities which are called persons*)
- $\exists \text{hasFriend.Person}$
 - (*Intuition: the set of entities which have – at least – one friend who is a person*)
- $\forall \text{hasFriend.Person}$
 - (*Intuition: the set of entities whose friends are only persons, possibly none*)
- Integer
 - (*Intuition: the set of Integers*)
- $\exists \text{hasApple.Integer}$
 - (*Intuition: the set of entities which have at least one apple*)
- $\forall \text{hasApple.Integer}$
 - (*Intuition: the set of entities which have only not halved apples, possibly none*)

Nested etype percepts – Example

- $\exists \text{talksTo.}(\exists \text{hasFriend.Person})$
 - **Intuition:** the set of entities which talk – at least once – to entities which have – at least – one friend who is a person
- $\forall \text{talksTo.}(\exists \text{hasFriend.Person})$
 - **Intuition:** the set of entities which talk only to the set of entities which have – at least – one friend who is a person
- $\exists \text{talksTo.}(\forall \text{hasFriend.Person})$
 - **Intuition:** the set of entities which talk – at least once – to the set of entities whose friends are only persons, possibly none
- $\forall \text{talksTo.}((\forall \text{hasFriend.Person})$
 - **Intuition:** the set of entities which talk only to the set of entities whose friends are only persons, possibly none

Example – how LoD (names of) percepts represent ETG facts



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Step 2 – The language of composite etype percepts

Definition (The language of composite etype percepts, L_C)

$$L_C = \langle A_C, FR_C \rangle = \{p_C\}$$

where:

- L_C is a **language** of composite etype percepts
- $A_C = \{p_T\}$, the **alphabet**, consists of all the percepts $p_T \in L_T$
- FR_C is a **set of formation rules**
- $\{p_C\}$ is the set of composite etype percepts a_C obtained by the exhaustive application of FR_C to A_C (the transitive closure $FR_C(A_C)$ of FR_C applied to A_C).

Formation rules – BNF

$$\begin{aligned} \langle p_C \rangle &::= \langle p_C \rangle \sqcap \langle p_C \rangle \mid \\ &\quad \langle p_C \rangle \sqcup \langle p_C \rangle \mid \\ &\quad \neg \langle p_C \rangle \\ \langle p_C \rangle &::= \langle p_T \rangle \end{aligned}$$

Formation rules – BNF

Notation (BNF). $\langle p_C \rangle$ is a nonterminal symbol and it stands for a p_C percept. $\langle p_T \rangle$ is an L_C terminal symbol and stands for an L_T percept. See above how the BNF of L_T expands to a LoD terminal symbol.

Observation (BNF). This BNF does allow the iterative application of the formation rules. It allows to generate percepts of any depth.

Observation (Conjunctive composite etypes). The intuition is that the conjunctive etype $\langle p_C \rangle \sqcap \langle p_C \rangle$ has the properties of both the component etypes. Semantically it means that composite its elements are elements of both the component etypes.

Observation (Disjunctive composite etypes). The intuition is that the disjunctive etype $\langle p_C \rangle \sqcup \langle p_C \rangle$ has the properties of at least one of the component etypes. Semantically it means that its elements are elements of one or both the component etype.

Observation (Complement composite etypes). The intuition is that the complement etype $\neg \langle p_C \rangle$ has the opposite properties of at least one of the component etypes. Semantically it means that its elements are all the elements of the universe of interpretation which are not elements of $\langle p_C \rangle$.

Observation (Composite etypes and world models). Composite etypes cannot be represented in the world models seen the past. This is the case in particular for disjunctive and complement etypes.

Composite etype percepts – Example

- $\text{Person} \sqcap \exists \text{hasFriend. Person}$
 - (***Intuition:** the set of entities which are persons and have a friend which is a person*)
- $\text{Person} \sqcup \text{Dog}$
 - (***Intuition:** the set of entities which are a person or a dog*)
- $\neg \exists \text{hasFriend. Person}$
 - (***Intuition:** the set of entities do not have a friend which is a person*)
- $\text{Person} \sqcap \neg (\exists \text{hasFriend. Person})$
 - (***Intuition:** the set of entities which are persons and which do not have a friend which is a person*)

Composite etype percepts – Example etypes

Consider the following concept names:

Vehicle, Boat, Bicycle, Car, Device, Wheel, Engine, Axle, Rotation,
Water, Human, Driver, Adult, Child

Formalize the following natural language statements:

- Nothing (empty set): $\perp$
- Everything (All the interpretation domain): $\top$
- Humans which are drivers : $\text{Human} \sqcap \text{Driver}$
- Humans and vehicles: $\text{Human} \sqcup \text{Vehicle}$
- Vehicles which are not boats: $\text{Vehicle} \sqcap \neg \text{Boat}$
- Wheels or engines which are used in cars: $(\text{Wheel} \sqcup \text{Engine}) \sqcap \text{Car}$
- Adults or children: $\text{Adult} \sqcup \text{Child}$

Composite etype percepts – Example roles

Consider the previous concept names plus the following role names:

hasPart, poweredBy, capableOf, travelsOn, controls

Formalize in DL the following natural language statements:

1. Those vehicles that have wheels and are powered by an engine
2. Those vehicles that have wheels and are powered by a human
3. Those vehicles that travel on water
4. Those objects which have no wheels
5. Those objects which do not travel on water
6. Those devices that have an axle and are capable of rotation
7. Those humans who control a vehicle
8. The drivers of cars

Composite etype percepts – Example roles (cont)

1. $\text{Vehicle} \sqcap \exists \text{hasPart.Wheel} \sqcap \exists \text{poweredBy.Engine}$
2. $\text{Vehicle} \sqcap \exists \text{hasPart.Wheel} \sqcap \exists \text{poweredBy.Human}$
3. $\text{Vehicle} \sqcap \exists \text{travelsOn.Water}$
4. $\forall \text{hasPart.} \neg \text{Wheel}$
5. $\forall \text{travelsOn.} \neg \text{Water}$
6. $\text{Device} \sqcap \exists \text{hasPart.Axle} \sqcap \exists \text{capableOf.Rotation}$
7. $\text{Human} \sqcap \exists \text{controls.Vehicle}$
8. $\text{Driver} \sqcap \exists \text{controls.Car}$

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Step 2 – The language of assertions

Definition (The language of descriptions, L_{LoD})

$$L_{LoD} = \langle A_{LoD}, FR_{LoD} \rangle = \{a_{LoD}\}$$

where:

- L_{LoD} is a **language of assertions**
- $A_{LoD} = \{a_c\}$, the **alphabet**, consists of all the formulas $a_c \in L_c$
- FR_{LoD} is a **set of formation rules**
- $\{a_{LoD}\}$ is the set of assertions a_{LoD} which are obtained by the exhaustive application of FR_{LoD} to A_{LoD} (the transitive closure $FR_{LoD}(A_{LoD})$ of FR_{LoD} applied to A_{LoD}). Each assertion a_{LoD} is called a **(LoD) assertion**.

(LoD) Descriptions – BNF

$$\begin{aligned} \langle a_{\text{LoD}} \rangle &::= \langle p_C \rangle \sqsubseteq \langle p_C \rangle \mid \\ &\quad \langle p_C \rangle \equiv \langle p_C \rangle \end{aligned}$$

where:

- $\langle p_C \rangle$ is a composite etype percept.
- $\langle a_{\text{LoD}} \rangle$ is a LoD description, an assertion involving two composite etype percepts.

(LoD) Descriptions – BNF

Observation (BNF). Differently from the previous two BNF's this BNF does not say how to build an etype from a simpler etype. Rather, it describes a **constraint** which must be satisfied by the two component etypes.

Terminology (LoD Description). A **LoD description** describes how the extensions of two composite etype percepts correlate. It is a constraint which reflects back into the component etypes. We call the first a **subsumption** (description) and the second an **equivalence** (description).

Terminology (Subsumption). $\sqsubseteq$ is a **subsumption relation**. $p_1 \sqsubseteq p_2$ is to be read as p_1 is **subsumed by** p_2 , or, vice versa that p_2 **subsumes** p_1 .

Terminology (Equivalence). $\equiv$ is an **equivalence relation**. We have

$$p_1 \equiv p_2 \text{ if and only if } p_1 \sqsubseteq p_2 \text{ and } p_2 \sqsubseteq p_1$$

Observation (LoD Description). Both types of descriptions state a «similarity» constraint, with subsumption being a weaker form of similarity

(LoD) Definitions – BNF

$$\langle a_{\text{LoD}} \rangle ::= \langle E \rangle \sqsubseteq \langle p_C \rangle \mid$$

$$\langle E \rangle \equiv \langle p_C \rangle$$

$$\langle E \rangle ::= E_1 \mid \dots \mid E_n$$

where:

- $\langle E \rangle$ is an atomic etype percept (an etype in L_T).
- $E_1 \mid \dots \mid E_n$ are (names of) etype percpcets
- p_C is a composite etype percept.
- a_{LoD} is a **LoD definition**.

(LoD) Definitions – BNF

Terminology (Etype subsumption, etype equivalence). As for LoD descriptions, the first definition is an **etype subsumption**. The second is an **etype equivalence**. Equivalences allow to precisely define the extension of $\langle E \rangle$.

Observation (LoD definition). A **LoD definition** is a LoD description that describes the extension of an atomic etype. It constrains the extension of $\langle E \rangle$. LoD definitions allow to introduce new etypes by defining their extension.

Observation (LoD definition). LoD definitions allow to extend indefinitely the language used to describe the world. This is a property used in natural language where, for instance, one uses the term «a friend» to mean «a person who is one's friend» or «driver» «a person who drives a car». LoD definitions extend to (formal and natural) languages the definitions which are usually used in mathematics.

Etype disjointness – Observation

Definition ((Atomic) etype disjointness): A special case of concept subsumption is **etype disjointness**, i.e.,

$$E_1 \sqsubseteq \neg E_2$$

where E_1 is an atomic etype and E_2 is an etype, possibly an atomic etype, also written

$$E_1 \perp E_2$$

If both E_1 and E_2 are atomic etypes, we have an **atomic etype disjointness**.

Observation (Etype disjointness): Strong etype disjointness, i.e.,

$$E_1 \equiv \neg E_2$$

written

$$E_1 \perp\!\!\!\perp E_2$$

implies that the union of the two etypes is the Universe of interpretation $U = \{u\}$.

Observation (Etype disjointness) Etype disjointness states that two etypes (e.g., “blue cars” and “yellow cars”) are disjoint.

Etype subsumption – Examples

1. Boats have no wheels
2. Cars do not travel on water
3. Drivers are adults who control cars
4. Humans are not vehicles
5. Wheels are not humans
6. Humans are either adults or children
7. Adults are not children

Etype subsumption – Examples

1. $\text{Boat} \sqsubseteq \forall \text{hasPart.} \neg \text{Wheel}$
2. $\text{Car} \sqsubseteq \forall \text{travelsOn.} \neg \text{Water}$
3. $\text{Driver} \sqsubseteq \text{Adult} \sqcap \exists \text{controls.Car}$
4. $\text{Human} \sqsubseteq \neg \text{Vehicle}$
5. $\text{Wheel} \sqsubseteq \neg \text{Human}$
6. $\text{Human} \sqsubseteq \text{Adult} \sqcup \text{Child}$
7. $\text{Adult} \sqsubseteq \neg \text{Child}$

Etype equivalence – Examples

1. Cars are exactly those vehicles that have wheels and are powered by an engine
2. Bicycles are exactly those vehicles that have wheels and are powered by a human
3. Boats are exactly those vehicles that travel on water
4. Wheels are exactly those devices that have an axle and are capable of rotation
5. Drivers are exactly those humans who control a vehicle

Etype equivalence – Examples

1. $\text{Car} \equiv \text{Vehicle} \sqcap \exists \text{hasPart.Wheel} \sqcap \exists \text{poweredBy.Engine}$
2. $\text{Bicycle} \equiv \text{Vehicle} \sqcap \exists \text{hasPart.Wheel} \sqcap \exists \text{poweredBy.Human}$
3. $\text{Boat} \equiv \text{Vehicle} \sqcap \exists \text{travelsOn.Water}$
4. $\text{Wheel} \equiv \text{Device} \sqcap \exists \text{hasPart.Axle} \sqcap \exists \text{capableOf.Rotation}$
5. $\text{Driver} \equiv \text{Human} \sqcap \exists \text{controls.Vehicle}$

LoD – The Logic of descriptions

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- Key notions

Interpretation function

Definition (Interpretation function, alphabet). The LoD interpretation function I is defined as

$$I : L \rightarrow D$$

where I is defined as

$$I = \langle I_T, I_C \rangle$$

where:

- I_T : the Type interpretation function,
- I_C : the complex type interpretation function

Interpretation function – Observations

Observation (Three LoD sub-languages composed in a single language). Each of the three languages generates, via appropriate formation rules, more complex formulas using the formulas generated by the previous language as elements of the alphabet.

Observation (Nested interpretation functions). Each language has its own interpretation function which uses the output of the interpretation function of the language one level below as its own input. We have nested interpretation functions.

Observation (Domain in extension via language constructs). The first two interpretation functions generate (formulas denoting) complex (etype and composite etype) percepts. There is no need for a third interpretation function for assertions.

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Interpretation of etype percepts

$I_T(T) = U$, with U the universe of interpretation

$I_T(\perp) = \emptyset$, with $\emptyset$ the empty set

$I_T(E_i) = E_i$

$I_T(D_i) = D_i$

$I_T(\exists P.T) = \{d \in U \mid \text{there is an } e \in U \text{ with } (d, e) \in I_T(P) \text{ and } e \in I_T(T)\}$

$I_T(\forall P.T) = \{d \in U \mid \text{for all } e \in U \text{ if } (d, e) \in I(P) \text{ then } e \in I_T(T)\}$

where I_T is the interpretation function of L_T

Observation (Interpretation function). For an intensional view of the interpretation functions for etypes, dtypes, object properties and attributes, follow what done with LoE.

Observation (Interpretation of nested etypes). It is sufficient to interpret the application of the second external quantifier to the etype built via the application of the first quantifier.

Interpretation of etype percepts - Observations

Observation (Existential type). $\exists P.T$ is an **existential etype**. Its interpretation

$$I(\exists P.T) = \{d \in U \mid \text{there is an } e \in U \text{ with } (d, e) \in I(P) \text{ and } e \in I(T)\}$$

is the set of all units d for which there exists a unit e in the codomain of P or type T . $\exists P.T$ defines the etype which is in relation P with T , not necessarily only with T (there could be an e' not in $I(T)$).

Example. $\exists \text{hasFriend.Person}$ is the etype of all those entities who have *at least* a friend who is a person.

Interpretation of etype percepts - Observations

Observation (Universal etype). $\forall P.T$ is an **universal etype**. Its interpretation

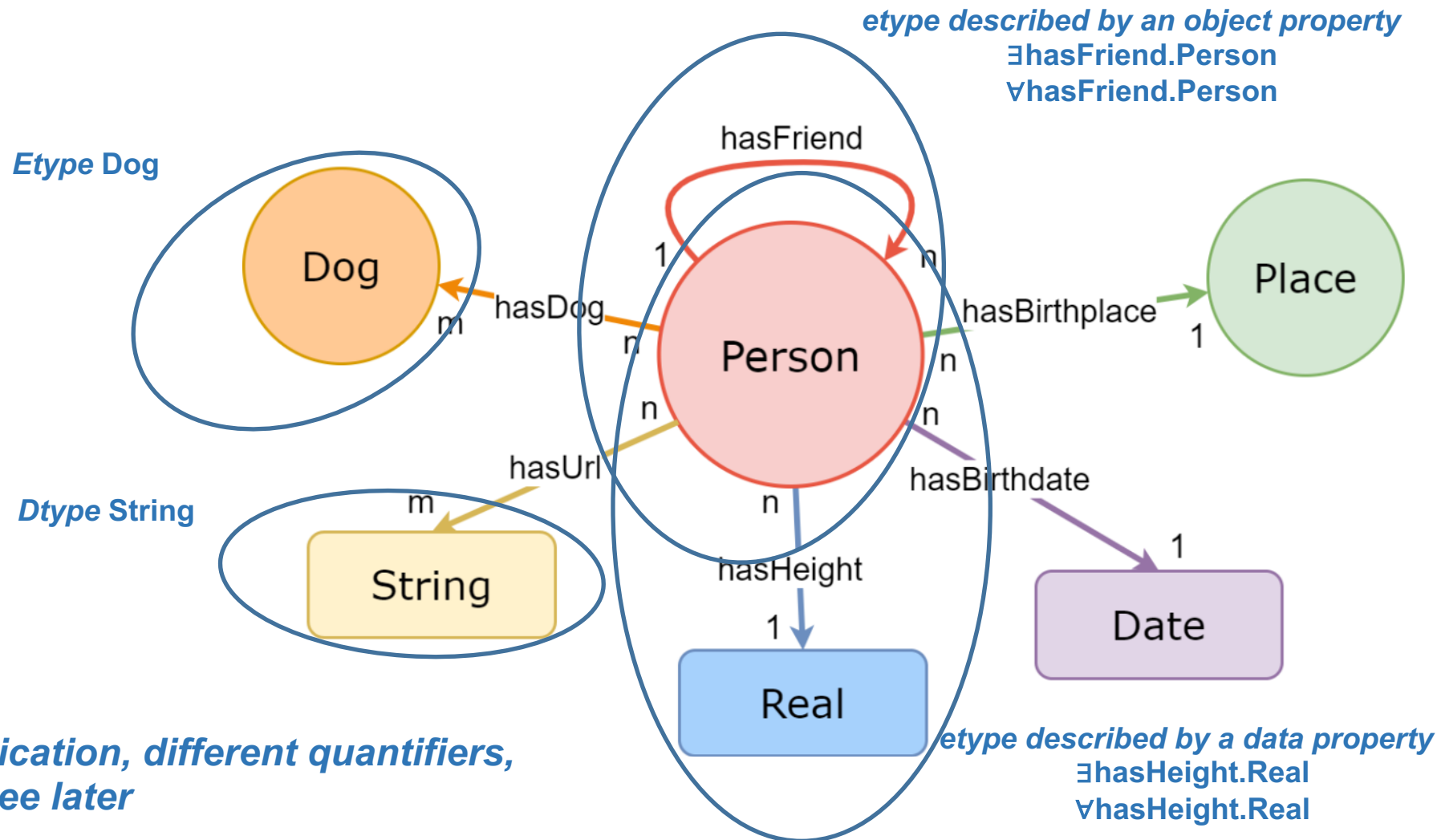
$$I(\forall P.T) = \{d \in U \mid \text{for all } e \in U \text{ if } (d, e) \in I(P) \text{ then } e \in I(T)\}$$

is the set of all units d for which all the units e in the codomain of P are of type T .

Example. $\forall \text{hasFriend.Person}$ is the etype of all people whose friends are *only* persons.

Proposition. $\forall P.T$ does not imply $\exists P.T$. $\forall P.T$ holds if $I(\forall P.T) = \emptyset$, while this is not the case with $\exists P.T$.

Example – how LoD (names of) percepts represent ETG facts



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Interpretation of composite etype percepts

$$I_C(p_1 \sqcap p_2) = I_C(p_1) \cap I_C(p_2)$$

$$I_C(p_1 \sqcup p_2) = I_C(p_1) \cup I_C(p_2)$$

$$I_C(\neg p_1) = U \setminus I_C(p_1)$$

$$I_C(p_T) = I_T(p_T)$$

$$I_T(p_T) = p_T$$

where:

- I_C is the interpretation function of L_C
- I_T is the interpretation function for L_T , the language of etype percepts.
- p_1, p_2 are composite etype percepts
- p_T (in *italic*) is (the name of an) etype percept denoting the domain percept p_T (not in *italic*)

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Interpretation of LoD descriptions

$$\begin{aligned} I(p_1 \sqsubseteq p_2) & \quad \text{iff} \quad I_c(p_1) \subseteq I_c(p_2) \\ I(p_1 \equiv p_2) & \quad \text{iff} \quad I_c(p_1) = I_c(p_2) \\ & \quad \text{iff} \quad I_c(p_1) \subseteq I_c(p_2) \text{ and } I_c(p_2) \subseteq I_c(p_1) \end{aligned}$$

where:

- I is the interpretation function for L_{LoD}
- I_c is the interpretation function for L_c , the language of composite etype percepts

Observation. Differently from before, the interpretation function does not return a subset of the domain but a fact.

Interpretation of LoD definitions

$$\begin{aligned} I(E \sqsubseteq p_2) & \quad \text{iff} \quad I_c(E) \subseteq I_c(p_2) \\ I(E \equiv p_2) & \quad \text{iff} \quad I_c(E) = I_c(p_2) \\ & \quad \text{iff} \quad I_c(E) \subseteq I_c(p_2) \text{ and } I_c(p_2) \subseteq I_c(E) \\ I_c(E) & \quad = \quad I_T(E) = E \end{aligned}$$

where:

- I is the interpretation function for L_{LoD}
- I_c is the interpretation function for L_c , the language of composite etype percepts
- I_T is the interpretation function for L_T , the language of composite etype percepts
- E is an etype, a subset of the universe of interpretation

Observation. Again, differently from before, the interpretation function does not return a subset of the domain but a fact.

Assertions and facts

Observation (LoD assertions). LoD assertions (descriptions and definitions) describe the following facts:

- A composite etype p_i is a subset of a composite etype p_j
- A composite etype p_i has the same extension of another composite etype p_j
- A (new atomic) etype E is a subset of a composite etype p
- A (new atomic) etype E has the same extension as a composite etype p .

Assertions and facts only have one of four possible forms:

- The assertion $p_i \sqsubseteq p_j$ describing the fact $p_i \subseteq p_j$
- The assertion $p_i \equiv p_j$ describing the fact $p_i = p_j$
- The assertion $E \sqsubseteq p$ describing the fact $E \subseteq p$
- The assertion $E \equiv p$ describing the fact $E = p$

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TBox – A LoD Theory

Definition (TBox). A LoD TBox is an assertional theory consisting of a set of etype LoD **descriptions** and / or etype **definitions**.

Observation (TBox). “T” in “TBox” stands for “Term” (LoD assertions correlate the extension of the terms describing the percepts).

Observation (Using TBoxes). TBoxes are used in DB and KR applications to make explicit the knowledge implicit in DBs and EGs.

(Strongly) definitional TBox

Definition (Definitional TBox) A LoD TBox is a finite set of LoD (etype) definitions.

Definition (Strongly definitional TBox) A LoD TBox is a finite set of LoD (etype) definitions which are **equivalences**.

Observation 4: Our focus is mainly on definitional TBoxes, which are key in AI in the modeling how human language and knowledge structure the world.

Strongly definitional TBOX (example)

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Organization}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \neg \text{Woman}$
- $\text{Mother} \equiv \text{Woman} \sqcap \exists \text{hasChild.Person}$
- $\text{Father} \equiv \text{Man} \sqcap \exists \text{hasChild.Person}$
- $\text{Parent} \equiv \text{Father} \sqcup \text{Mother}$

LoD Terminology

Definition (Uses). Let T be definitional TBox. Let $E \sqsubseteq p$ or $E \equiv p$ be a definition in T . Then we say that E **directly uses** E' , where E' is an atomic etype, if E' occurs in p . We say that E **uses** E' if E' occurs in the right hand side of a definition of an etype mentioned in p , and so on recursively.

Observation (Uses). “Uses” is defined as the transitive closure of directly uses.

Observation (Acyclic definitional TBox). A definitional TBox is **acyclic** if

- There is no type that uses itself, and
- There are no two definitions of the same etype

Observation 1 (acyclic TBox). The second requirement avoids any type using itself.

Observation 2 (acyclic TBox). An acyclic Tbox avoids the following situation:

$$E_1 \sqsubseteq \dots E_2 \dots, \quad E_2 \sqsubseteq \dots E_3 \dots, \quad \dots, \quad E_n \sqsubseteq \dots E_1 \dots$$

Definition (Terminology). A **Terminology** is an acyclic definitional TBox.

Observation (Terminology). Terminologies are key in the construction of human knowledge.

Two terminologies (example) – trees rooted in a etype

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$

Family relations

- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \equiv \text{Parent} \sqcap \text{Female}$
- $\text{Father} \equiv \text{Parent} \sqcap \text{Male}$

Composed terminology (not a great example)

Family relations

- $\text{Person} \equiv \text{Man} \sqcup \text{Woman}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \neg \text{Woman}$
- $\text{Mother} \equiv \text{Woman} \sqcap \exists \text{hasChild}.\text{Person}$
- $\text{Father} \equiv \text{Man} \sqcap \exists \text{hasChild}.\text{Person}$
- $\text{Parent} \equiv \text{Father} \sqcup \text{Mother}$

Composed terminology (better – still not a great example)

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \neg \text{Woman}$
- $\text{Mother} \equiv \text{Woman} \sqcap \exists \text{hasChild.Person}$
- $\text{Father} \equiv \text{Man} \sqcap \exists \text{hasChild.Person}$
- $\text{Parent} \equiv \text{Father} \sqcup \text{Mother}$

Composed terminology (a good example)

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \equiv \text{Parent} \sqcap \text{Female}$
- $\text{Father} \equiv \text{Parent} \sqcap \text{Male}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$

Composed terminology (a good example)

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Female} \sqperp \text{Male}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \sqsubseteq \text{Parent}$
- $\text{Father} \sqsubseteq \text{Parent}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$

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World models as etype graphs (Recall)

Etype graphs are linguistic representations of analogical representations which **describe** etypes. They are **graphs** where

- Nodes are etypes (as in «married person»)
- There is any number of link denoting **object** and **data properties**, (as in «friend of», «has friend», «has height», the last cannot be reversed)

where: etypes and properties denote subsets of the percepts and pairs or percepts, respectively, as denoted by a **reference concept** which defines the universe of interpretation

NOTE: etype graphs may or may not have «**more/less general (MG/LG) etypes**» or **etype disjointness**» relations denoting, respectively, subset and empty intersection.

Etype graphs – Types of knowledge models (Recall)

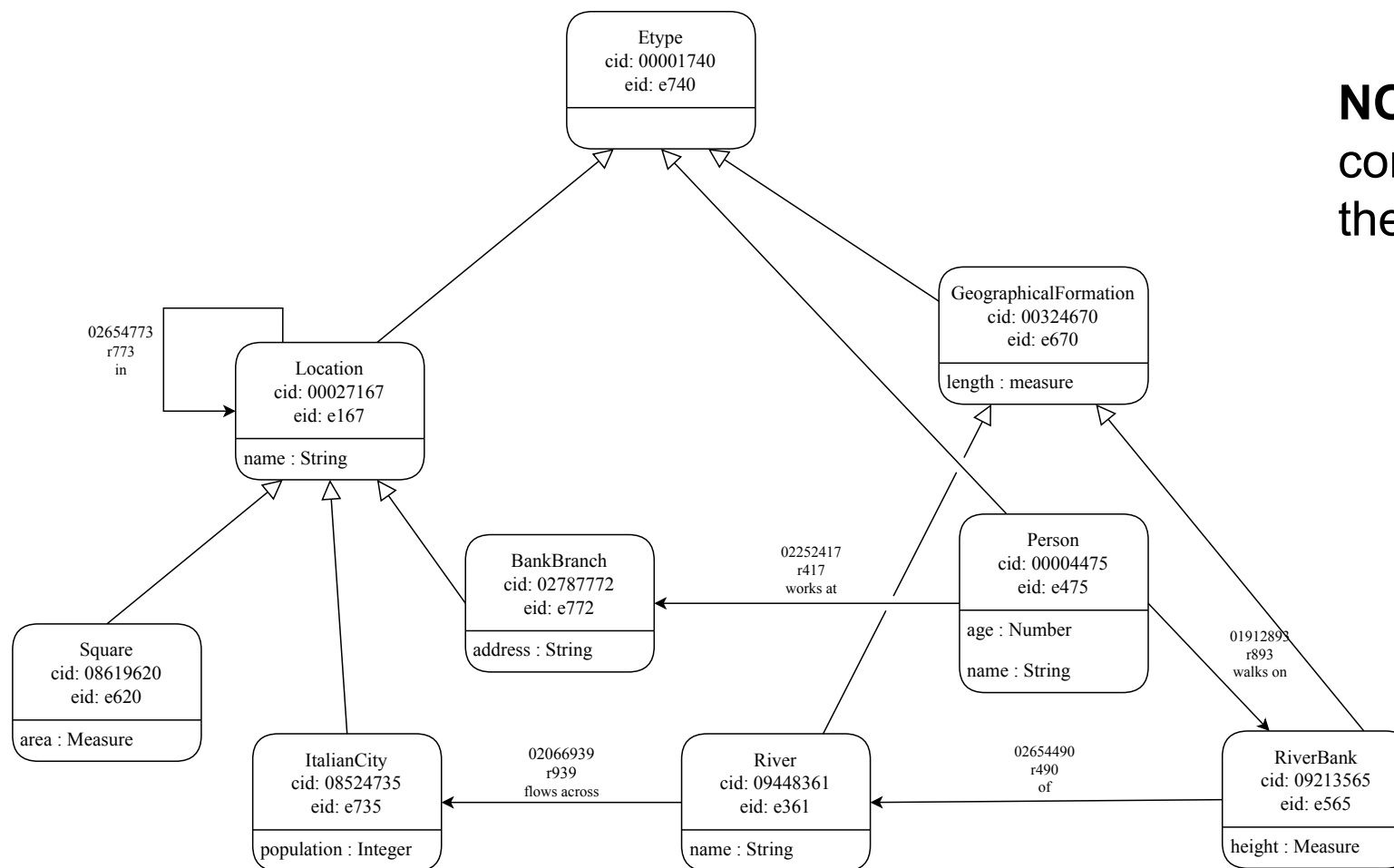
Terminology (Knowledge ontologies). Knowledge ontologies are knowledge models constructed from ontologies where one or more concepts are refined into etypes.

Terminology (Teleontologies, taxonomies). (Knowledge) teleontologies (KTLO) are knowledge ontologies with one or more MG/LG relations. **Taxonomies (KTAX)** are tree-like KTLO where the extension of the union of the children is the same as the extension of the parent node.

Observation (knowledge ontologies). Similarly to language models and ontologies, knowledge ontologies are distinguished based on the types of concepts and the generality of the top level concept (e.g., **top level KTLO, domain KTLO, task KTLO and application KTLO**).

Terminology (Teleologies). Teleologies (TLO) are KTLO with MG/LG relation only to the root etype (i.e., reference concept).

Knowledge teleontology (KTLO) – example (recall)

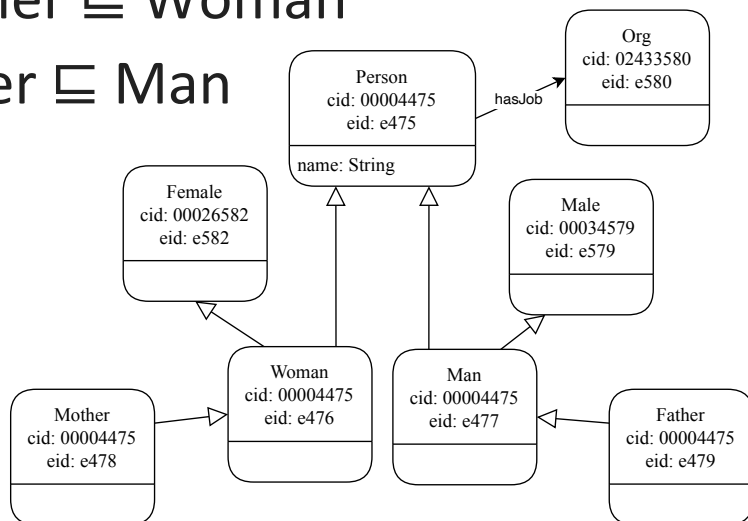


NOTE: Cid's of types connect to the Cid of the reference concept

Two terminologies (example) – DAGs rooted in 3 etypes

Family relations

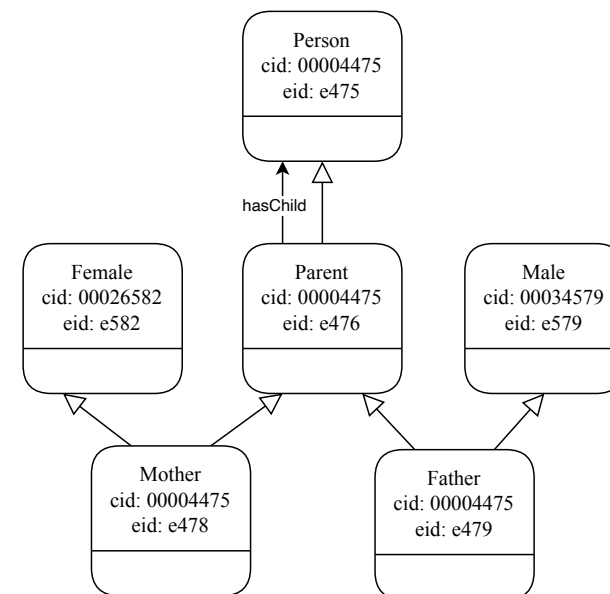
- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$



Family relations

- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \equiv \text{Parent} \sqcap \text{Female}$
- $\text{Father} \equiv \text{Parent} \sqcap \text{Male}$

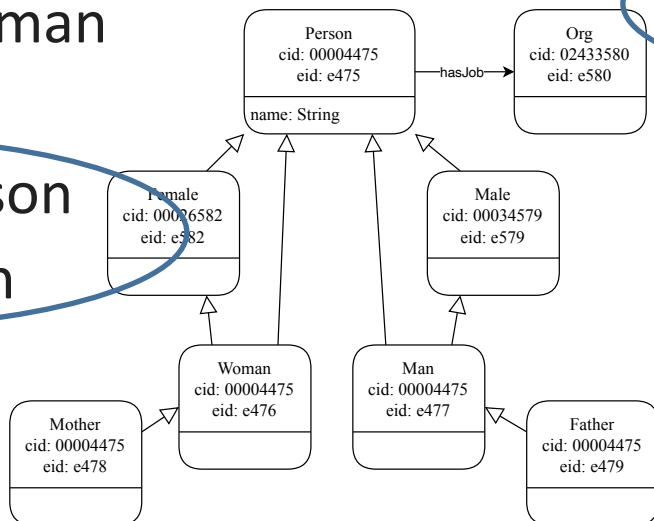
**NOT a
teleontology!
Neither of them.**



Two KTLOs (example) – DAGs rooted in 1 etype

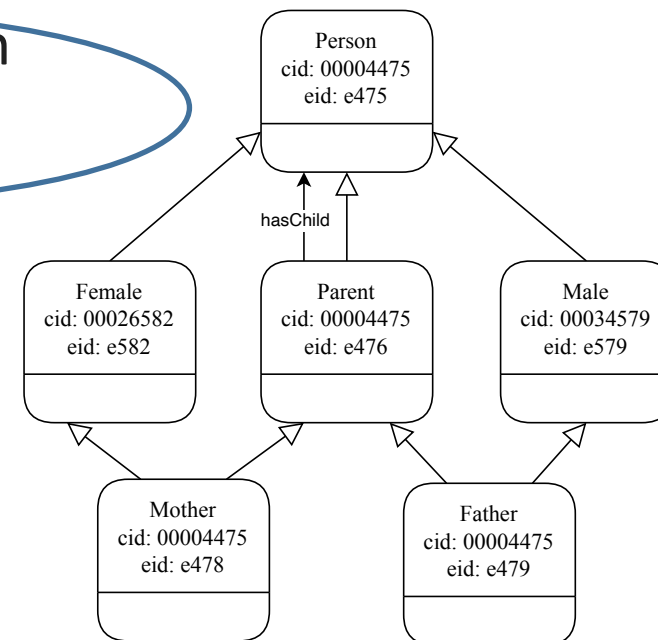
Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$
- $\text{Female} \sqsubseteq \text{Person}$
- $\text{Male} \sqsubseteq \text{Person}$



Family relations

- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \equiv \text{Parent} \sqcap \text{Female}$
- $\text{Father} \equiv \text{Parent} \sqcap \text{Male}$
- $\text{Female} \sqsubseteq \text{Person}$
- $\text{Male} \sqsubseteq \text{Person}$



Merge 2 into 1 terminology (not a great example)

Family relations

- $\text{Person} \equiv \text{Man} \sqcup \text{Woman}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \neg \text{Woman}$
- $\text{Mother} \equiv \text{Woman} \sqcap \exists \text{hasChild}.\text{Person}$
- $\text{Father} \equiv \text{Man} \sqcap \exists \text{hasChild}.\text{Person}$
- $\text{Parent} \equiv \text{Father} \sqcup \text{Mother}$

**Not a World Model
representation!**

Merge 2 into 1 terminology (better – still not a great example)

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \neg \text{Woman}$
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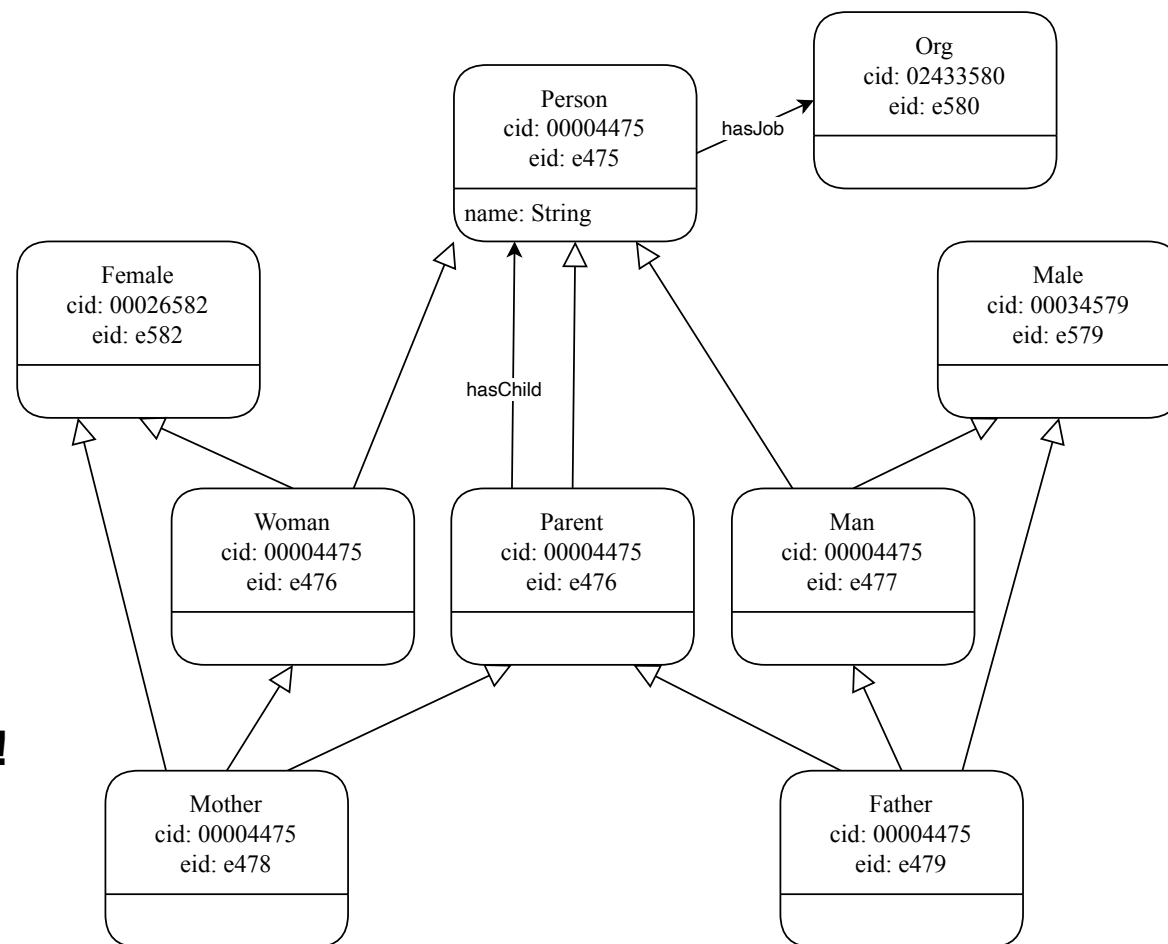
**Not a World Model
representation!**

Merge 2 into 1 terminology (example) – a DAG rooted in 4 etypes

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \equiv \text{Parent} \sqcap \text{Female}$
- $\text{Father} \equiv \text{Parent} \sqcap \text{Male}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$

**NOT a
teleontology!**

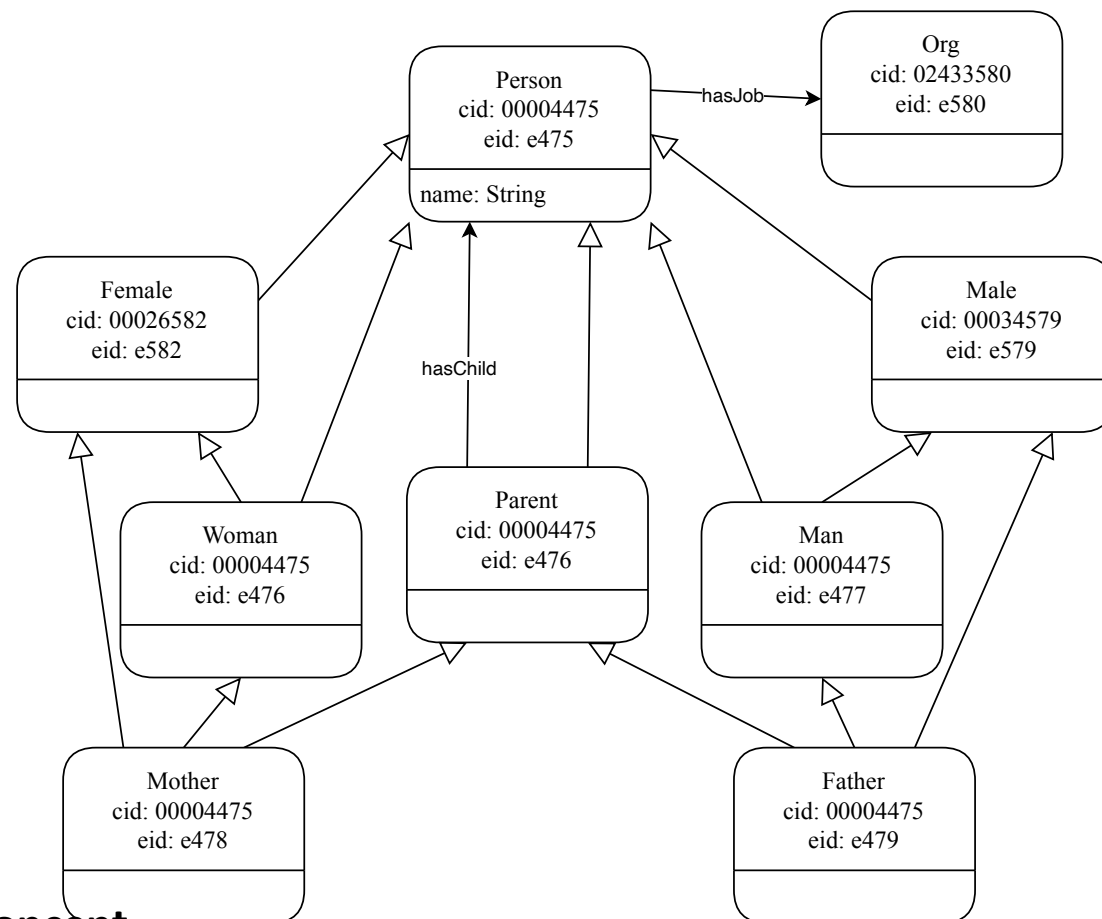


Merge 2 into 1 KTLOs (example) – a DAG rooted in a etype

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \equiv \text{Parent} \sqcap \text{Female}$
- $\text{Father} \equiv \text{Parent} \sqcap \text{Male}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$
- $\text{Female} \sqsubseteq \text{Person}$
- $\text{Male} \sqsubseteq \text{Person}$

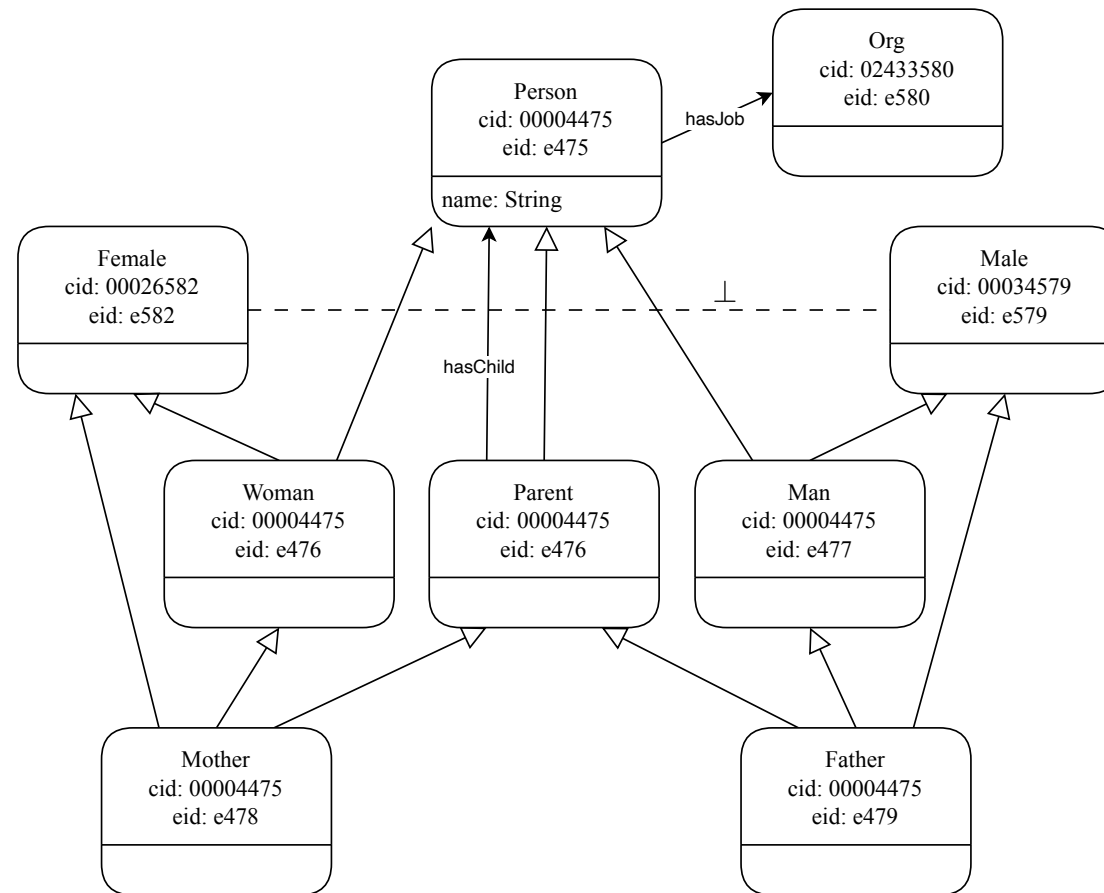
Under the assumption that Org is a concept datatype (e.g. a type of string)



Composed terminology (example) – a DAG rooted in a etype

Family relations

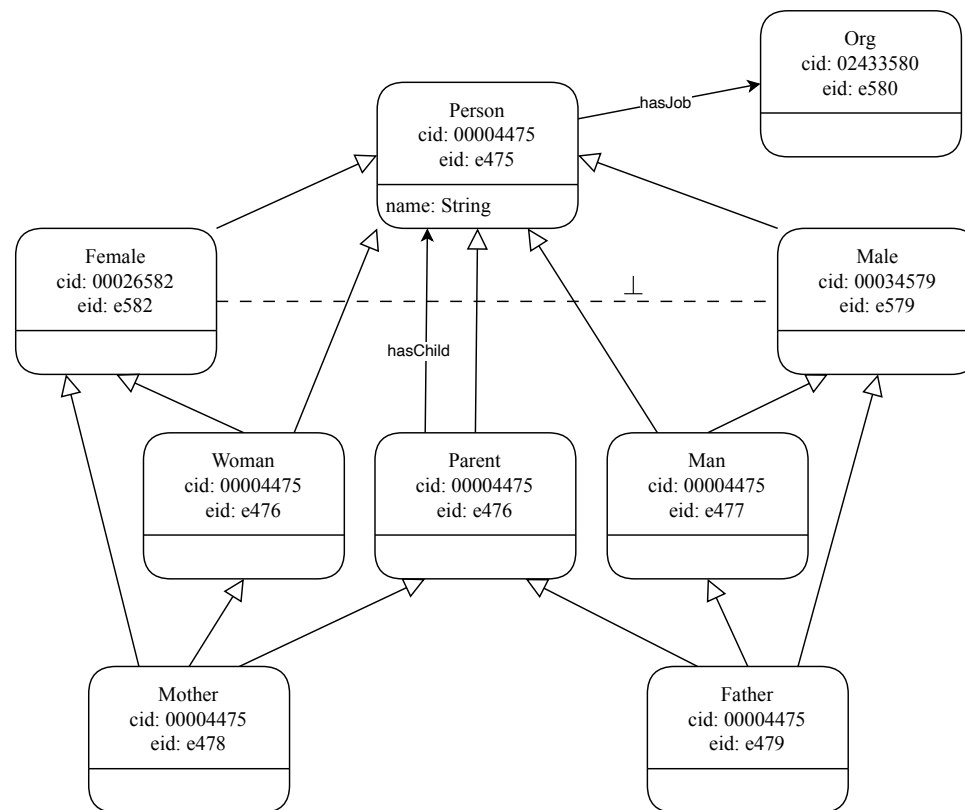
- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Female} \perp \text{Male}$
- $\text{Woman} \equiv \text{Person} \sqcap \text{Female}$
- $\text{Man} \equiv \text{Person} \sqcap \text{Male}$
- $\text{Parent} \equiv \text{Person} \sqcap \exists \text{hasChild.Person}$
- $\text{Mother} \sqsubseteq \text{Parent}$
- $\text{Father} \sqsubseteq \text{Parent}$
- $\text{Mother} \sqsubseteq \text{Woman}$
- $\text{Father} \sqsubseteq \text{Man}$



Composed KTLO (example) – a DAG rooted in a etype

Family relations

- $\text{Person} \equiv \exists \text{hasname.String} \sqcap \forall \text{HasJob.Org}$
- $\text{Female} \perp \text{Male}$
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- $\text{Father} \sqsubseteq \text{Man}$
- $\text{Female} \sqsubseteq \text{Person}$
- $\text{Male} \sqsubseteq \text{Person}$



Proposition (Terminology, teleontology). A teleontology is a type of terminology with constraints (e.g., one root).

LoD – The Logic of descriptions

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Etype graphs – Types of knowledge models (Recall)

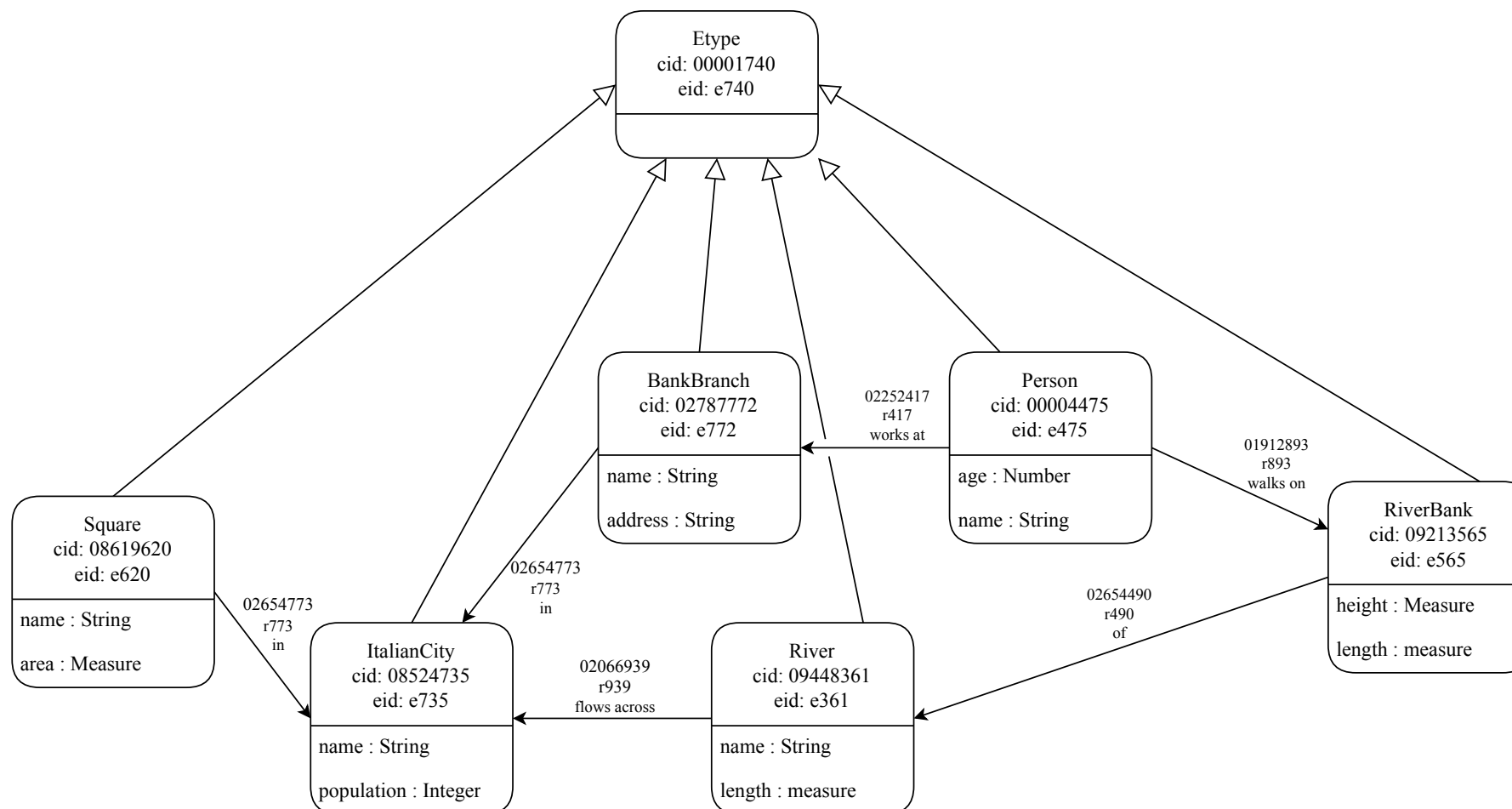
Terminology (Knowledge ontologies). Knowledge ontologies are knowledge models constructed from ontologies where one or more concepts are refined into etypes.

Terminology (Teleontologies, taxonomies). (Knowledge) teleontologies (KTLO) are knowledge ontologies with one or more MG/LG relations. **Taxonomies (KTAX)** are tree-like KTLO where the extension of the union of the children is the same as the extension of the parent node.

Observation (knowledge ontologies). Similarly to language models and ontologies, knowledge ontologies are distinguished based on the types of concepts and the generality of the top level concept (e.g., **top level KTLO, domain KTLO, task KTLO and application KTLO**).

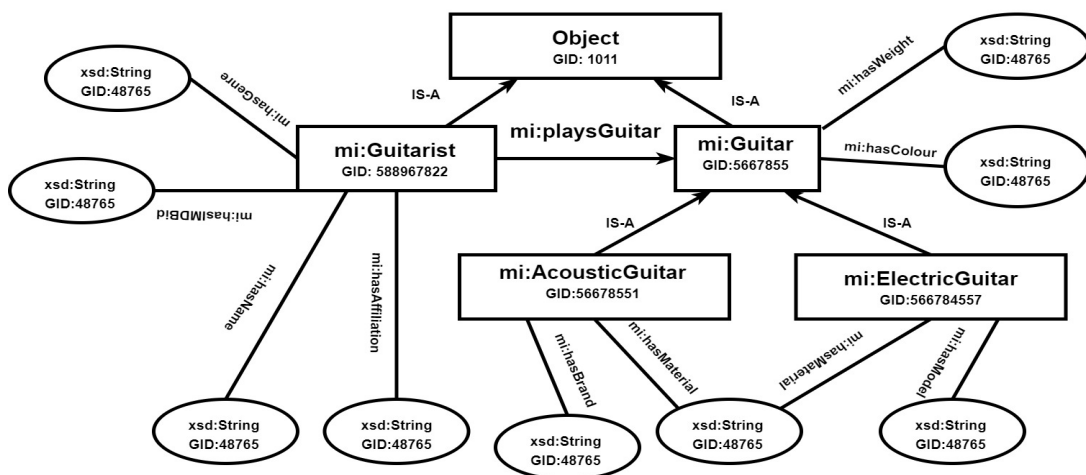
Terminology (Teleologies). Teleologies (TLO) are KTLO with MG/LG relation only to the root etype (i.e., reference concept).

Teleology (TLO) – example (Recall)

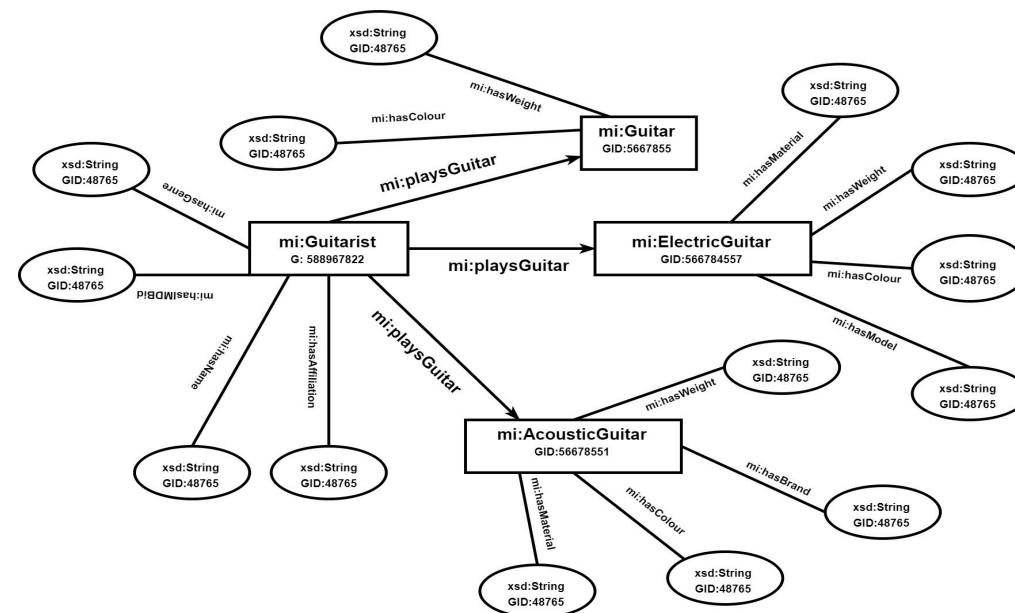


UNFOLDING - from KTLO to TLO – example (Recall)

FROM



TO



NOTE: Unfolding is an inference rule of the world logic of ETGs (so called LoD) which allows to unfold a KTLO into a LTLO by **selectively making explicit** the knowledge implicit in MG/LG links. It allows to flatten the KTLO by distributing properties to more specific etypes. Unfolding will be formalized when introducing LoD.

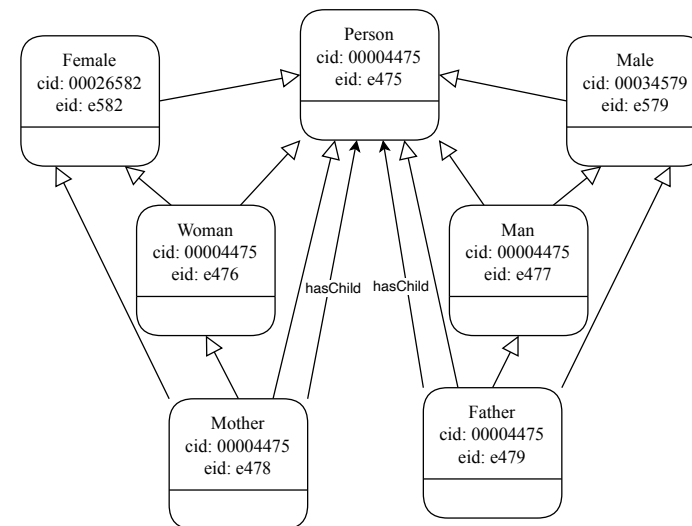
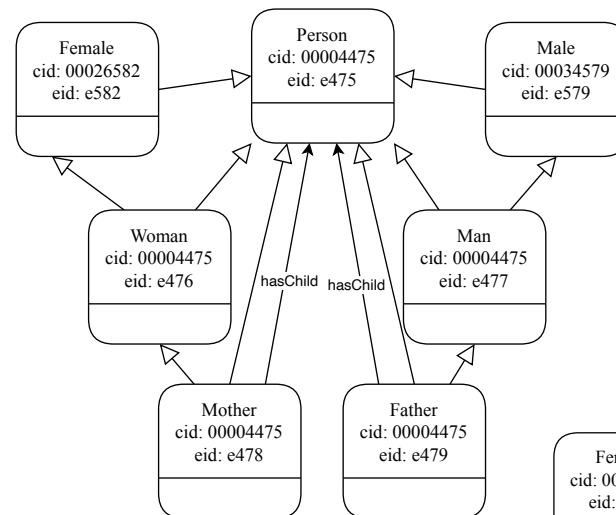
From teleontologies to teleologies – example

FROM

$Woman \equiv Person \sqcap Female$
 $Man \equiv Person \sqcap Male$
 $Mother \equiv Woman \sqcap \exists HasChild.Person$
 $Father \equiv Man \sqcap \exists HasChild.Person$
 $Male \sqsubseteq Person$
 $Female \sqsubseteq Person$

TO

$Woman \equiv Person \sqcap Female$
 $Man \equiv Person \sqcap Male$
 $Mother \equiv Person \sqcap Female \sqcap \exists HasChild.Person$
 $Father \equiv Person \sqcap Male \sqcap \exists HasChild.Person$
 $Male \sqsubseteq Person$
 $Female \sqsubseteq Person$



NOTE: controlled **unfolding** via reasoning

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World entailment (Recall)

Definition (World entailment, satisfiability) Let $W = \langle L_a, D, I_a \rangle$ be a world model. Let L_a be an assertional language. Then $\models_{L_a}$ is a **world entailment relation**

$$\models_{L_a} \subseteq D \times L_a$$

such that

$$M \models_{L_a} T_a$$

$M = \{f\} \subseteq D$ is a **model of** $T_a \subseteq L_a$. We say that M **entails** T_a . We write $M \models T_a$ instead of $M \models_{L_a} T_a$ when no confusion arises.

LoD world entailment

$$M \models p_1 \sqsubseteq p_2 \quad \text{iff} \quad I(p_1 \sqsubseteq p_2) \quad \text{iff} \quad I(p_1) \subseteq I(p_2)$$

$$\begin{aligned} M \models p_1 \equiv p_2 \quad \text{iff} \quad I(p_1 \equiv p_2) \quad \text{iff} \quad I(p_1) = I(p_2) \\ \text{iff} \quad I(p_1) \subseteq I(p_2) \text{ and } I(p_2) \subseteq I(p_1) \end{aligned}$$

with $p_1, p_2 \in L_{\text{LoD}}$.

World entailment

Observation (LoD world entailment). LoD world entailment reduces to the LoD interpretation function I_{LoD} . Namely an interpretation function I_{LoD} is a model M if and only if the assertion question a_{LoD} is a description or definition which states a property which is true in $M = I_{LoD}$

Observation (LoD World entailment). LoD world entailment checks for **equality** and **similarity**. The intuition is that the LoD language allows for complex assertions (both definitions and descriptions). The LoD assertion question amount to checking whether two percepts, constructed in different way actually name the same set or a smaller set.

Reminder (LoE world entailment). LoE entailment reduces to the test of whether $I_{LoE}(a_{LoE})$ is an element of the model $M = I_{LoE}$

Observation (LoD vs. LoE World entailment). Both LoD and LoE world entailment reduce to the interpretation function. LoD entailment is more complex than LoE entailment because the LoD language allows for more complex assertions.



World entailment (extended question language)

$$M \models p_1 \sqsubseteq p_2 \quad \text{iff} \quad I(p_1 \sqsubseteq p_2) \quad \text{iff} \quad I(p_1) \subseteq I(p_2)$$

$$M \models p_1 \equiv p_2 \quad \text{iff} \quad I(p_1 \equiv p_2) \quad \text{iff} \quad I(p_1) = I(p_2) \\ \text{iff} \quad I(p_1) \subseteq I(p_2) \text{ and } I(p_2) \subseteq I(p_1)$$

$$M \models p_1 \perp p_2 \quad \text{iff} \quad I(p_1 \sqsubseteq \neg p_2) \quad \text{iff} \quad I(p_1) \cap I(p_2) \subseteq \emptyset$$

$$M \models p_1 \perp\!\!\!\perp p_2 \quad \text{iff} \quad I(p_1 \equiv \neg p_2) \quad \text{iff} \quad I(p_1) \cap I(p_2) \subseteq \emptyset \text{ and } I(p_1) \cup I(p_2) \subseteq U$$

with

- $p_1, p_2 \in L_{\text{LoD}}$;
- $p_1 \perp p_2$ a notational variant of $p_1 \sqsubseteq \neg p_2$.
- $p_1 \perp\!\!\!\perp p_2$ a notational variant of $p_1 \equiv \neg p_2$.



World entailment (extended question language)

Observation (World entailment). World entailment can be extended to check for **equality / inequality** and **similarity / dissimilarity**.

Observation (Entailment language): The entailment (query) language can be extended / changed as long as it is possible to perform the translation into the underlying logic language in **polynomial time**. The underlying algorithm remains the same.

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Logical entailment (Recall)

Definition (Logical entailment). Let $L_W = \langle W, \models_W \rangle$ be a world logic with $W = \langle L, D, I \rangle$ its world model. Let $\{M\} \subseteq D$ be a set of models of T with $T, \underline{I} \subseteq L$ and $a \in L$. Then we write

$$T \models_{\{M\}} \underline{I} \qquad (T \models_{\{M\}} a)$$

and say that T **logically entails** $\underline{I}$ (T **logically entails** a) with respect to $\{M\}$ if

$$\text{for all } M \in \{M\}, \text{ if } M \models T \text{ then } M \models \underline{I} \text{ (} M \models a \text{)}$$

Observation (Model filtering via logical entailment). Logical entailment implements reasoning in the following way: “Whatever we derive from T must be true in the same sets of “relevant” models $\{M\}$ as T . T acts as a filter which allows to discard all the irrelevant models.

Observation (Model filtering via selection of $\{M\}$). The choice of the models $\{M\}$ of T is arbitrary. And it also amounts to model filtering. It could be the only model of T , all models of T , a model selected because it is the intended model of T , and so on. If $M \not\models T$ then Logical entailment does not apply. T has the key role of selecting which models $M \in \{M\}$ must be considered.

Notation ($\{M\}$). When no confusion arises we write $\models$ instead of $\models_{\{M\}}$.

LoD – Reasoning problems

Observation (LoD reasoning problems). The four LoD core reasoning problems are:

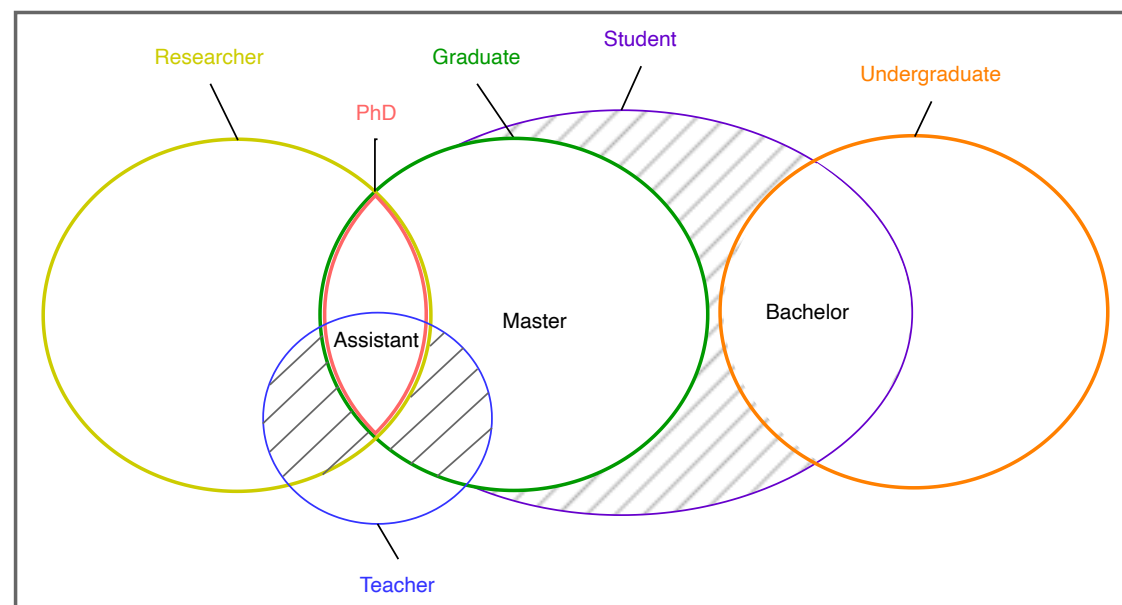
- $T \models C$ Satisfiability with respect to a TBox T
- $T \models C \sqsubseteq D$ Subsumption with respect to a TBox T
- $T \models C \equiv D$ Equivalence with respect to a TBox T
- $T \models C \perp D$ Disjointness with respect to a TBox T

where T can also be empty.

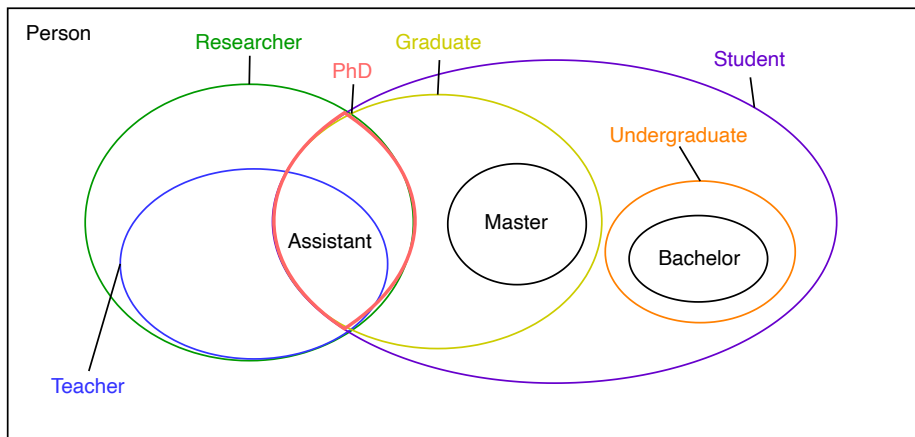
Observation (Empty TBox). All LoD reasoning problems are logical entailment problems. When the TBox is empty, then LoD logical entailment reduces to LoD world entailment on the reference model(s).

A terminology – running example

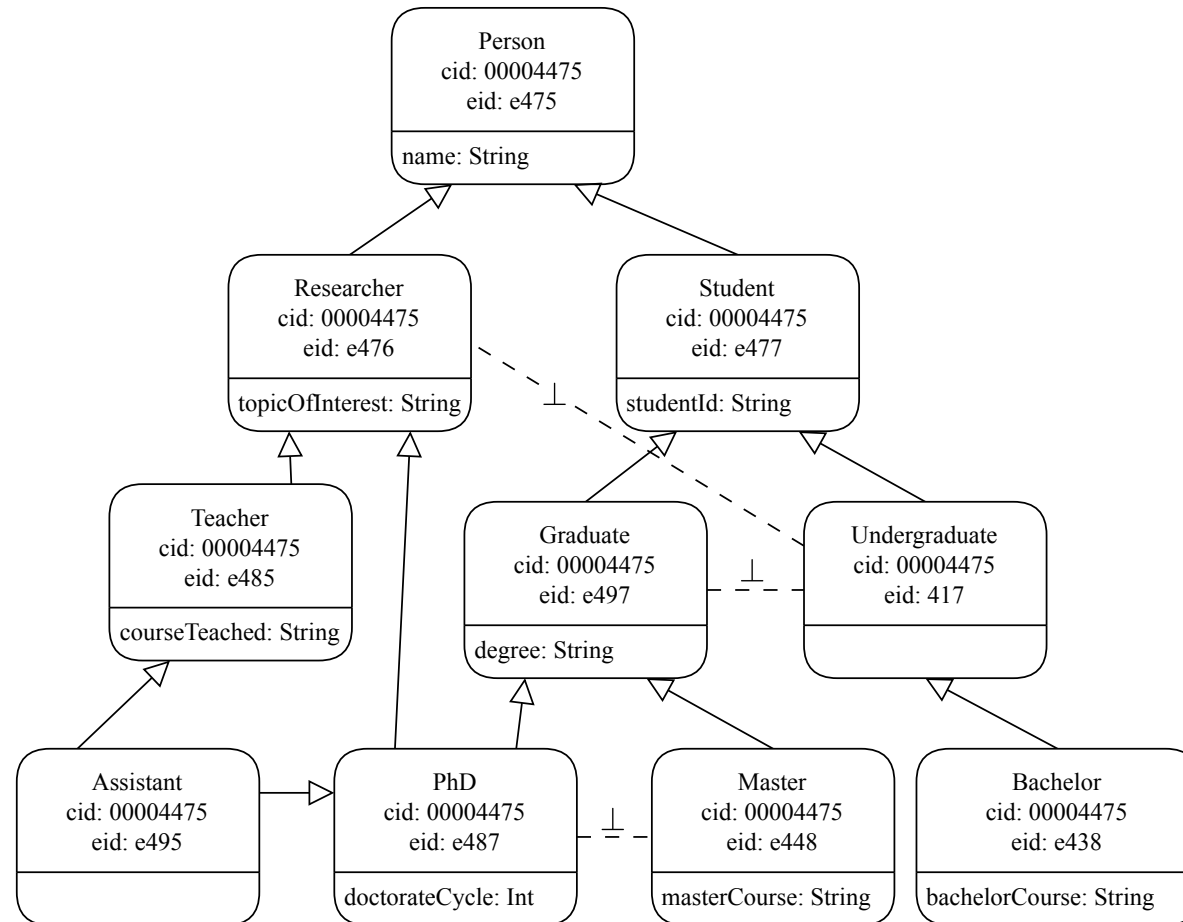
$$T = \begin{cases} \text{Student} \equiv \text{PhD} \sqcup \text{Master} \sqcup \text{Bachelor} \\ \text{Master} \equiv \text{Graduate} \sqcap \neg \text{Researcher} \\ \text{PhD} \equiv \text{Graduate} \sqcap \text{Researcher} \\ \text{Assistant} \equiv \text{PhD} \sqcap \text{Teacher} \\ \text{Bachelor} \sqsubseteq \text{Undergraduate} \\ \text{Graduate} \perp \text{Undergraduate} \end{cases}$$



A teleontology – running example

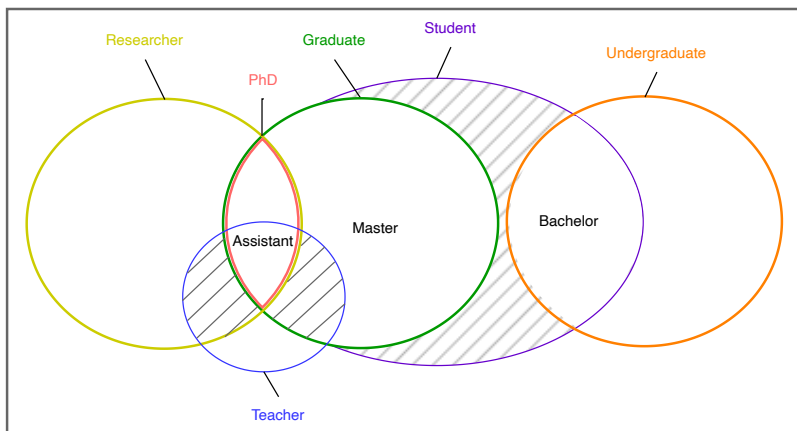


$$T = \left\{ \begin{array}{l} \text{Researcher} \sqsubseteq \text{Person} \\ \text{Student} \sqsubseteq \text{Person} \\ \text{Graduate} \sqsubseteq \text{Student} \\ \text{Undergraduate} \sqsubseteq \text{Student} \\ \text{Master} \sqsubseteq \text{Graduate} \\ \text{Bachelor} \sqsubseteq \text{Undergraduate} \\ \text{Teacher} \sqsubseteq \text{Researcher} \\ \text{PhD} \equiv \text{Graduate} \sqcap \text{Researcher} \\ \text{Assistant} \equiv \text{PhD} \sqcap \text{Teacher} \\ \text{Graduate} \perp \text{Undergraduate} \\ \text{PhD} \perp \text{Master} \\ \text{Undergraduate} \perp \text{Researcher} \end{array} \right.$$

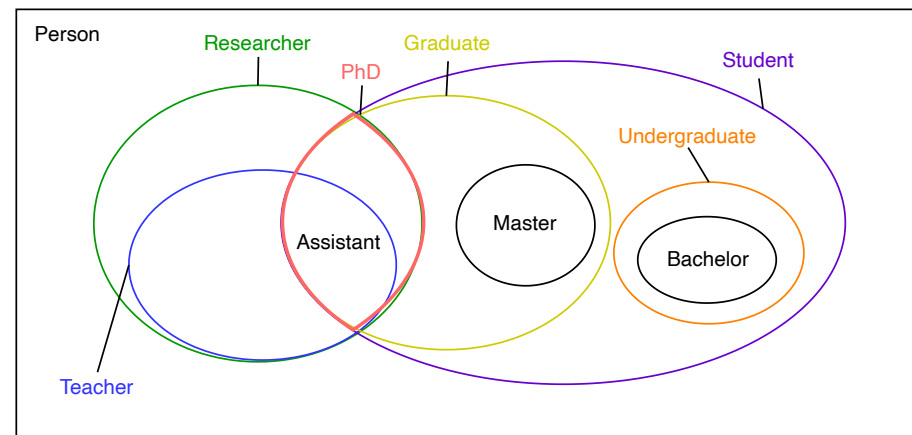


A terminology and a teleontology – a comparison

$$T = \begin{cases} \text{Student} \equiv \text{PhD} \sqcup \text{Master} \sqcup \text{Bachelor} \\ \text{Master} \equiv \text{Graduate} \sqcap \neg \text{Researcher} \\ \text{PhD} \equiv \text{Graduate} \sqcap \text{Researcher} \\ \text{Assistant} \equiv \text{PhD} \sqcap \text{Teacher} \\ \text{Bachelor} \sqsubseteq \text{Undergraduate} \\ \text{Graduate} \perp \text{Undergraduate} \end{cases}$$



$$T = \begin{cases} \text{Researcher} \sqsubseteq \text{Person} \\ \text{Student} \sqsubseteq \text{Person} \\ \text{Graduate} \sqsubseteq \text{Student} \\ \text{Undergraduate} \sqsubseteq \text{Student} \\ \text{Master} \sqsubseteq \text{Graduate} \\ \text{Bachelor} \sqsubseteq \text{Undergraduate} \\ \text{Teacher} \sqsubseteq \text{Researcher} \\ \text{PhD} \equiv \text{Graduate} \sqcap \text{Researcher} \\ \text{Assistant} \equiv \text{PhD} \sqcap \text{Teacher} \\ \text{Graduate} \perp \text{Undergraduate} \\ \text{PhD} \perp \text{Master} \\ \text{Undergraduate} \perp \text{Researcher} \end{cases}$$



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Satisfiability

$T \models C$ Satisfiability with respect to a TBox T ,
where C is an etype

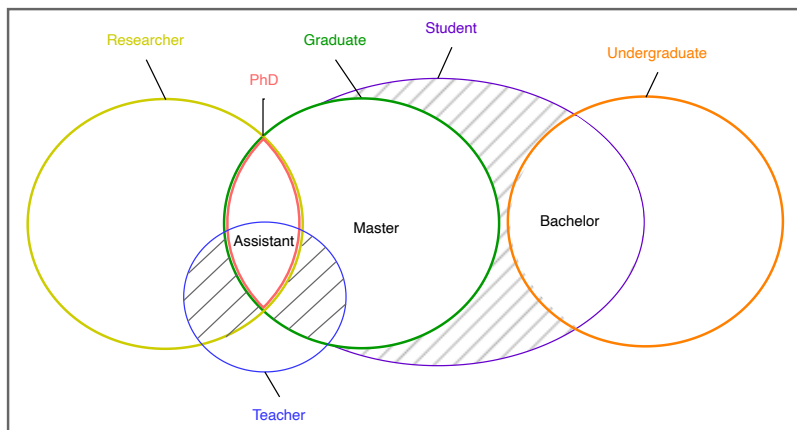
Definition (Satisfiability with respect to a TBox T). An etype C is **satisfiable** with respect to a TBox T if there exists a model I of T such that $I(C)$ is not empty. If it is not satisfiable, then C is said to be **unsatisfiable**.

Observation. A model is an interpretation function such that $I(C)$ is not empty

Terminology satisfiability – via Venn diagrams

$$T = \begin{cases} Student \equiv PhD \sqcup Master \sqcup Bachelor \\ Master \equiv Graduate \sqcap \neg Researcher \\ PhD \equiv Graduate \sqcap Researcher \\ Assistant \equiv PhD \sqcap Teacher \\ Bachelor \sqsubseteq Undergraduate \\ Graduate \perp Undergraduate \end{cases}$$

$$T \models Bachelor \sqcap PhD ?$$



NOTE: the diagram on the right is only one of the many possible ones.

To prove the goal one would have to generate **all** the possible diagrams till the satisfying interpretation function has been found, or all of them have been tried out. In the latter case the input formula is not satisfiable.

How to generate all possible diagrams?

Terminology satisfiability – via entailment

$$T \models \textit{Bachelor} \sqcap \textit{PhD} ? \textbf{(No)}$$

Proof:

$$\begin{aligned} & \textit{Bachelor} \sqcap \textit{PhD} \\ & \sqsubseteq \textit{Undergraduate} \sqcap (\textit{Graduate} \sqcap \textit{Researcher}) \\ & \sqsubseteq \textit{Undergraduate} \sqcap (\neg \textit{Undergraduate} \sqcap \textit{Researcher}) \\ & \equiv \textbf{\textit{Undergraduate}} \sqcap \neg \textbf{\textit{Undergraduate}} \sqcap \textit{Researcher} \\ & \equiv \perp \sqcap \textit{Researcher} \\ & \equiv \perp \end{aligned}$$

$$T = \begin{cases} \textit{Student} \equiv \textit{PhD} \sqcup \textit{Master} \sqcup \textit{Bachelor} \\ \textit{Master} \equiv \textit{Graduate} \sqcap \neg \textit{Researcher} \\ \textit{PhD} \equiv \textit{Graduate} \sqcap \textit{Researcher} \\ \textit{Assistant} \equiv \textit{PhD} \sqcap \textit{Teacher} \\ \textit{Bachelor} \sqsubseteq \textit{Undergraduate} \\ \textit{Graduate} \perp \textit{Undergraduate} \end{cases}$$

Observation (Entailment). In this proof, and all the following, the reasoning strategy is to use definitions to expand formulas as long as needed. This process, called **unfolding**, is that implemented by the decision procedure informally introduced earlier on and formally defined later

Teleontology satisfiability – via entailment

$T \models \textit{Bachelor} \sqcap \textit{PhD} ? \textbf{(No)}$

Proof:

$\textit{Bachelor} \sqcap \textit{PhD}$

$\sqsubseteq \textit{Undergraduate} \sqcap (\textit{Graduate} \sqcap \textit{Researcher})$

$\sqsubseteq \textit{Undergraduate} \sqcap \neg \textit{Undergraduate} \sqcap \textit{Researcher}$

$\equiv \perp \sqcap \textit{Researcher}$

$\equiv \perp$

$T = \left\{ \begin{array}{l} \textit{Researcher} \sqsubseteq \textit{Person} \\ \textit{Student} \sqsubseteq \textit{Person} \\ \textit{Graduate} \sqsubseteq \textit{Student} \\ \textit{Undergraduate} \sqsubseteq \textit{Student} \\ \textit{Master} \sqsubseteq \textit{Graduate} \\ \textit{Bachelor} \sqsubseteq \textit{Undergraduate} \\ \textit{Teacher} \sqsubseteq \textit{Researcher} \\ \textit{PhD} \equiv \textit{Graduate} \sqcap \textit{Researcher} \\ \textit{Assistant} \equiv \textit{PhD} \sqcap \textit{Teacher} \\ \textit{Graduate} \perp \textit{Undergraduate} \\ \textit{PhD} \perp \textit{Master} \\ \textit{Undergraduate} \perp \textit{Researcher} \end{array} \right.$

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Subsumption

$T \models C \sqsubseteq D$ Subsumption with respect to a TBox T

Definition (Subsumption with respect to a TBox T). An assertion C is subsumed by an assertion D with respect to a TBox T if

$$I(C) \subseteq I(D)$$

for every model I of T . In which case we also write

$$C \sqsubseteq_T D$$

Terminology subsumption – via entailment

$$\mathcal{T} \models \textit{Student} \sqsubseteq \textit{Graduate} \sqcup \textit{Undergraduate} ? \text{ (Yes)}$$
$$\mathcal{T} = \begin{cases} \textit{Student} \equiv \textit{PhD} \sqcup \textit{Master} \sqcup \textit{Bachelor} \\ \textit{Master} \equiv \textit{Graduate} \sqcap \neg \textit{Researcher} \\ \textit{PhD} \equiv \textit{Graduate} \sqcap \textit{Researcher} \\ \textit{Assistant} \equiv \textit{PhD} \sqcap \textit{Teacher} \\ \textit{Bachelor} \sqsubseteq \textit{Undergraduate} \\ \textit{Graduate} \perp \textit{Undergraduate} \end{cases}$$

Proof:

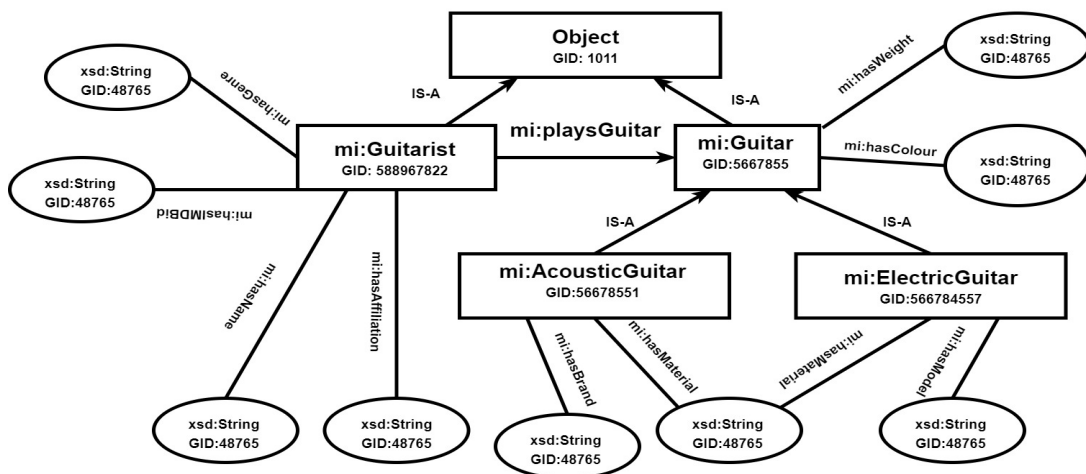
$$\begin{aligned} \textit{Student} &\equiv \textit{PhD} \sqcup \textit{Master} \sqcup \textit{Bachelor} \\ &\equiv (\textit{Graduate} \sqcap \textit{Researcher}) \sqcup (\textit{Graduate} \sqcap \neg \textit{Researcher}) \sqcup \textit{Bachelor} \\ &\sqsubseteq (\textit{Graduate} \sqcap \textit{Researcher}) \sqcup (\textit{Graduate} \sqcap \neg \textit{Researcher}) \sqcup \textit{Undergraduate} \\ &\sqsubseteq \textit{Graduate} \sqcup \textit{Undergraduate} \end{aligned}$$

The union of two non-empty sets is not empty.

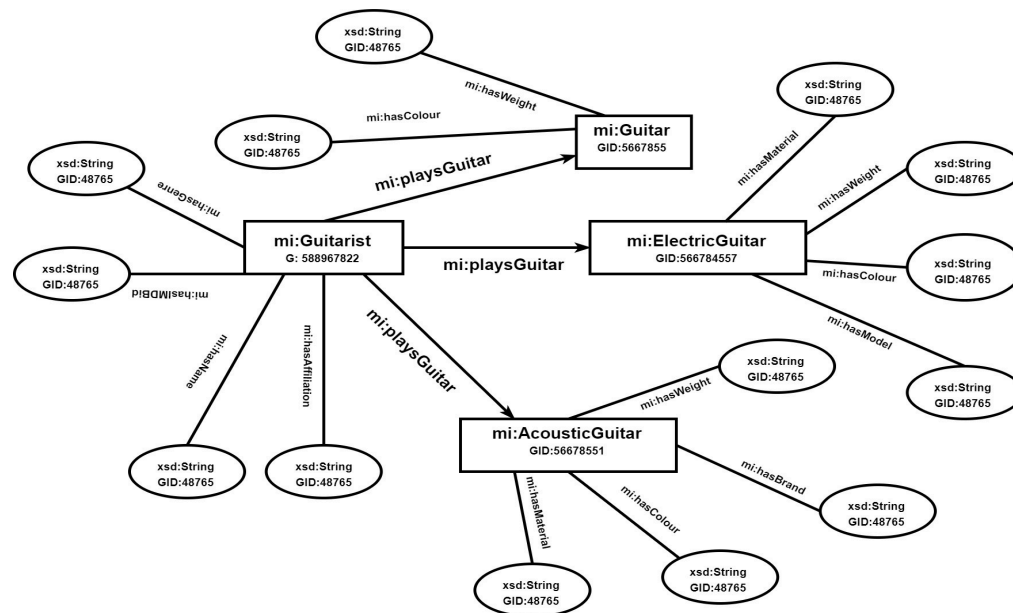
Observation (Entailment). The last step is the result of the (implicit exponential) expansion of the disjunction over the conjunction. Disjunctions create decision points on which disjunct should be expanded first. This does not happen with conjunctions as both conjuncts need to be satisfied.

UNFOLDING - from KTLO to TLO – example (Recall)

FROM



TO



NOTE: Unfolding is an inference rule of the world logic of ETGs (so called LoD) which allows to unfold a KTLO into a LTLO by **selectively making explicit** the knowledge implicit in MG/LG links. It allows to flatten the KTLO by distributing properties to more specific etypes. Unfolding will be formalized when introducing LoD.

Teleontology subsumption – via entailment

$T \models PhD \sqsubseteq Student$? (Yes)

Proof:

PhD

$\equiv Researcher \sqcap Graduate$

$\sqsubseteq Researcher \sqcap Student$

$\sqsubseteq Student$

$$T = \left\{ \begin{array}{l} Researcher \sqsubseteq Person \\ Student \sqsubseteq Person \\ Graduate \sqsubseteq Student \\ Undergraduate \sqsubseteq Student \\ Master \sqsubseteq Graduate \\ Bachelor \sqsubseteq Undergraduate \\ Teacher \sqsubseteq Researcher \\ PhD \equiv Graduate \sqcap Researcher \\ Assistant \equiv PhD \sqcap Teacher \\ Graduate \perp Undergraduate \\ PhD \perp Master \\ Undergraduate \perp Researcher \end{array} \right.$$

Observation (Entailment). Reasoning with teleontologies cannot generate exponential growth of the formula as there is no disjunction and negation /disjointness applies to atomic etypes (the negation of a conjunction is a conjunction).

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Equivalence

$$T \models C \equiv D$$

Equivalence with respect to a TBox T

Definition(Equivalence with respect to a TBox T). Two assertions C and D are equivalent with respect to a TBox T if

$$I(C) = I(D)$$

for every model I of T . In this case we also write

$$C \equiv_T D$$

Terminology equivalence – via entailment

$$T = \begin{cases} Student \equiv PhD \sqcup Master \sqcup Bachelor \\ Master \equiv Graduate \sqcap \neg Researcher \\ PhD \equiv Graduate \sqcap Researcher \\ Assistant \equiv PhD \sqcap Teacher \\ Bachelor \sqsubseteq Undergraduate \\ Graduate \perp Undergraduate \end{cases}$$

$T \models Graduate \equiv Phd \sqcup Master?$ **(Yes)**

Proof:

$$\begin{aligned} & PhD \sqcup Master \\ & \equiv (Graduate \sqcap Researcher) \sqcup (Graduate \sqcap \neg Researcher) \\ & \equiv Graduate \sqcup (Researcher \sqcap \neg Researcher) \\ & \equiv Graduate \end{aligned}$$

Teleontology equivalence – via entailment

$T \models \text{Assistant} \equiv \text{Graduate} \sqcap \text{Researcher} \sqcap \text{Teacher?}$ (Yes)

Proof:

Assistant
 $\equiv \text{PhD} \sqcap \text{Teacher}$
 $\equiv \text{Graduate} \sqcap \text{Researcher} \sqcap \text{Teacher}$

$T = \left\{ \begin{array}{l} \text{Researcher} \sqsubseteq \text{Person} \\ \text{Student} \sqsubseteq \text{Person} \\ \text{Graduate} \sqsubseteq \text{Student} \\ \text{Undergraduate} \sqsubseteq \text{Student} \\ \text{Master} \sqsubseteq \text{Graduate} \\ \text{Bachelor} \sqsubseteq \text{Undergraduate} \\ \text{Teacher} \sqsubseteq \text{Researcher} \\ \text{PhD} \equiv \text{Graduate} \sqcap \text{Researcher} \\ \text{Assistant} \equiv \text{PhD} \sqcap \text{Teacher} \\ \text{Graduate} \perp \text{Undergraduate} \\ \text{PhD} \perp \text{Master} \\ \text{Undergraduate} \perp \text{Researcher} \end{array} \right.$

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 - Subsumption
 - Equivalence
 - **Disjointness**
 - Reduction to Satisfiability
- Key notions

Disjointness

$$T \models C \perp D$$

Disjointness with respect to a TBox T

Definition (Disjointness with respect to a TBox T). Two assertions C and D are disjoint with respect to T if

$$I(C) \cap I(D) = \emptyset$$

for every model I of T . In this case we also write

$$C \perp_T D$$

Terminology disjointness – via entailment

$T \models \text{Assistant} \sqcap \text{Master} \sqsubseteq \perp$? (**Yes**)

Proof:

$$\begin{aligned} & \text{Assistant} \sqcap \text{Master} \\ & \equiv (\text{PhD} \sqcap \text{Researcher}) \sqcap (\text{Graduate} \sqcap \neg \text{Researcher}) \\ & \equiv \perp \sqcap \text{PhD} \sqcap \text{Graduate} \\ & \sqsubseteq \perp \end{aligned}$$

$$T = \begin{cases} \text{Student} \equiv \text{PhD} \sqcup \text{Master} \sqcup \text{Bachelor} \\ \text{Master} \equiv \text{Graduate} \sqcap \neg \text{Researcher} \\ \text{PhD} \equiv \text{Graduate} \sqcap \text{Researcher} \\ \text{Assistant} \equiv \text{PhD} \sqcap \text{Teacher} \\ \text{Bachelor} \sqsubseteq \text{Undergraduate} \\ \text{Graduate} \perp \text{Undergraduate} \end{cases}$$



Teleontology disjointness – via entailment

$T \models \text{Undergraduate} \sqcap \text{Assistant} \sqsubseteq \perp ?$ (Yes)

Proof:

$\text{Undergraduate} \sqcap \text{Assistant}$
 $\sqsubseteq \neg \text{Researcher} \sqcap \text{Assistant}$
 $\equiv \neg \text{Researcher} \sqcap (\text{PhD} \sqcap \text{Teach})$
 $\sqsubseteq \neg \text{Researcher} \sqcap \text{PhD} \sqcap \text{Researcher}$
 $\equiv \perp \sqcap \text{PhD}$
 $\equiv \perp$

$T = \left\{ \begin{array}{l} \text{Researcher} \sqsubseteq \text{Person} \\ \text{Student} \sqsubseteq \text{Person} \\ \text{Graduate} \sqsubseteq \text{Student} \\ \text{Undergraduate} \sqsubseteq \text{Student} \\ \text{Master} \sqsubseteq \text{Graduate} \\ \text{Bachelor} \sqsubseteq \text{Undergraduate} \\ \text{Teacher} \sqsubseteq \text{Researcher} \\ \text{PhD} \equiv \text{Graduate} \sqcap \text{Researcher} \\ \text{Assistant} \equiv \text{PhD} \sqcap \text{Teacher} \\ \text{Graduate} \perp \text{Undergraduate} \\ \text{PhD} \perp \text{Master} \\ \text{Undergraduate} \perp \text{Researcher} \end{array} \right.$

LoD – The Logic of descriptions

- Intuition
- Definition
- Domain
- Language
 - Etype percepts
 - Composite etype percepts
 - Descriptions / definitions
- Interpretation function
 - Etype percepts
 - Composite etype percepts
 - Descriptions / definitions
- Theories
 - Teleontologies
 - Teleologies
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- Key notions

Reasoning problems (Reduction)

Proposition (Reduction to satisfiability). All the problems reduce to satisfiability. In fact, we have the following equivalences:

- **Equivalence:** $C \equiv_T D$ if and only if $C \sqsubseteq_T D$ and $D \sqsubseteq_T C$;
- **Subsumption:** $C \sqsubseteq_T D$ if and only if $C \sqcap \neg D$ is *unsatisfiable* with respect to T ;
- **Disjointness:** $C \perp_T D$ if and only if $C \sqcap D$ is *unsatisfiable* with respect to T .

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Key notions

- LoD as a logic of etypes
- Etype graphs (ETGs)
- Language of etype percepts
- Nested etype percepts
- Language of composite etype percepts
- LoD assertions
- LoD facts
- Language of descriptions
- Descriptions
- Definitions
- Entailment
- Tell – TBoxes
- (Strongly) definitional TBox
- Acyclic TBox
- Terminology
- Satisfiability
- Subsumption
- Equivalence
- Disjointness
- Reasoning problem reducibility



LoD

The logic of Descriptions

(HP2T)